

A bridge to definiteness: Identifying unique and relational definites through bridging

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Abstract

Definiteness covers a range of related meanings such as uniqueness and familiarity that are difficult to tease apart. The notion of bridging provides a testing ground for identifying the underlying mechanisms licensing definite expressions. In this paper, we investigate the extent to which Mandarin bare nouns and demonstratives license unique and relational bridging, exploring how these expressions carve up the definite space and differ from English definite expressions. Through comprehension and production studies, we show that Mandarin bare nouns and demonstratives readily allow unique and relational bridging, while English demonstratives are degraded. We further show that bare nouns license relational bridging only if the bridged noun is lexically relational, while demonstratives do not have such a restriction. We analyze Mandarin demonstrative *na* as a relationalizing operator that introduces an external index and a relational predicate in the restriction. This study has implications on existing analyses of Mandarin definite expressions and linguistic markings of definiteness, relationality, and anaphora.

1 Introduction

Definiteness is a concept that is difficult to define. While the term ‘definite determiner’ morphosyntactically maps onto the determiner *the* in languages like English, the meanings associated with *the* include a range of notions from existence, uniqueness, familiarity, and more (Coppock and Beaver 2015; Heim 1983; Roberts 2003; Schwarz 2009; Wespel 2008; a.o.). Notions such as uniqueness and familiarity are often difficult to tease apart, as illustrated in examples like (1).

- (1) I met a linguist yesterday. The linguist talked about bridging.

In (1), the referent of ‘the linguist’ can be resolved either through situational uniqueness — because the referent is unique in the minimal situation described — or through familiarity — because the indefinite expression in the first sentence makes a relevant referent available for an anaphoric pickup. Many languages have been shown to distinguish the two mechanisms morphosyntactically (Schwarz 2009, 2013, a.o.), but because familiarity always entails uniqueness in some restricted domain, the boundary between the two notions is not clearly distinguished.

In this paper, we investigate a specific use of definite expressions, known as bridging, to further probe the type of definiteness that underlies a given expression. While the term ‘bridging’ covers a range of phenomena that is not confined to definiteness (see Clark 1975 and below for further discussion), we focus on the notion that is used in the literature looking at the licensing conditions of definite expressions. In this literature, the term refers to the phenomenon of a definite expression being licensed despite not having its referent introduced in the given context. When there is no clear referent available to resolve the definite expression to, speakers can make use of various strategies to resolve it for a coherent discourse, and these strategies are constrained by general pragmatic principles as well as lexical meanings of the bridged words (Asher and Lascarides 1998; Matsui

1995). For example, in (2) the referent of *the speaker* in the second sentence is not overtly given in the context, but is easily resolved by its relation to previous discourse: either through its unique existence in the situation described in the first sentence, or through its one-to-one relation to an entity introduced there (*the talk*). If the second sentence is presented without the first, it would be degraded due to a lack of a suitable referent for the definite description. We use the term ‘bridging’ in this paper to refer to the range of links that can be drawn between the referent of a definite description and some property of the context that licenses it.

(2) I attended a talk yesterday. The speaker talked about bridging.

Different bridges can be built depending on the different relations linking the previous discourse and the entity denoted by the definite expression. This means that, by manipulating the kinds of contexts in which definite descriptions appear, we can determine which semantic mechanism licenses the definite expression. This has been illustrated for languages like German, where two kinds of meanings underlying definiteness — uniqueness and familiarity — are morphosyntactically distinguished (Schwarz 2009). Schwarz (2009) shows that, in German, the uniqueness-denoting definite form shows up when the bridging link between the definite description and the context is that of uniqueness, and the familiarity-denoting definite form shows up when the bridging link is that of relational anaphora. By looking at bridging cases and separating the antecedent and the referent of the definite expression, we can more clearly tease apart anaphoric reference from situational uniqueness.

We apply this strategy and present a case study with Mandarin, whose inventory of definite expressions differs greatly from that of English and German. Mandarin lacks a definite article and marks definiteness using bare nouns and demonstrative descriptions (Cheng and Sybesma 1999; Jiang 2012), and how they map onto the uniqueness/familiarity distinction of definiteness has been debated in the recent literature (Bremmers et al. 2022; Dayal and Jiang 2021; Jenks 2018; a.o.). Specifically, Jenks (2018) argues that the different meanings associated with the English article *the* are divided between bare nouns and demonstratives in Mandarin, while Dayal and Jiang (2021) argue that Mandarin demonstratives should have an interpretation that is separate from that of *the*. By testing uniqueness- and familiarity-based bridging with Mandarin bare nouns and demonstratives, we reveal the mechanisms of definiteness underlying these expressions.¹ Through this strategy, we make several novel observations. First, we show that both bare nouns and demonstrative descriptions in Mandarin readily license uniqueness-based bridging and familiarity-based bridging (Study 1 and 3), in contrast to English where only definite descriptions, and not demonstrative descriptions, allow the two kinds of bridging (Study 2). Based on these observations, we propose that Mandarin demonstrative *na* contributes a possessive relation function that can take a relatum argument. This analysis predicts that when the noun is inherently non-relational and does not come with its own internal relational argument, only demonstratives, and not bare nouns, would license familiarity-based bridging. We show that this prediction is borne out (Study 4).

This study has several empirical and theoretical implications. As far as the authors know,

¹A reviewer asks how the different mechanisms of definiteness can be tested if Mandarin bare nouns allow non-definite interpretations. In our study, we restricted all of our stimuli so that only the definite interpretations would be available. First, all of the contexts we use in our studies involve uniqueness or familiarity, and we know that in such contexts, an indefinite expression is blocked by the definite counterpart by principles like *Maximize Presupposition!* (Heim 1991) in English. Using the same logic, we assume that the bare noun receives a definite reading even if an indefinite interpretation is available. Second, the literature on bare nouns that make use of typeshifting argue that the definite interpretation from the ι operator takes precedence over the indefinite interpretation from the $\exists$ operator (Chierchia 1998b; Dayal 2004; a.o.).

bridging in Mandarin has not been tested systematically in previous works, so this study serves as an empirical basis for determining which definite expressions allow which kinds of bridging in Mandarin. Our data show that Mandarin bare nouns and demonstratives allow both uniqueness- and familiarity-based bridging, which is not straightforwardly accounted for by previous works and calls for a modified analysis. Theoretically, our analysis has implications for analyses of demonstratives and relational nouns. Across languages, demonstratives are assumed to disallow bridging altogether, and so the unavailability of bridging uses has been used as a diagnostic for identifying demonstratives from definite descriptions in typological studies (Himmelman 1996; see Grosz 2018 for the bridging use of German demonstrative pronouns). Our study shows that an expression commonly identified as a demonstrative, one that displays all of the other properties used to identify demonstratives such as allowing deixis and anaphora, can still allow bridging, therefore suggesting that bridging is not a categorical diagnostic for distinguishing definites from demonstratives across languages. For relational nouns, our analysis of *na* implies there being two positions that can introduce the relatum argument: inside the noun denotation for inherently relational nouns, and outside the NP for non-relational nouns. We show that this aligns with the lexical and the pragmatic relational meanings identified in other works (Barker 2011; Vikner and Jensen 2002) and links the latter use to the use of demonstratives or other external operators that contribute a relational interpretation even if the noun does not come with a lexical argument slot.

The rest of the paper is organized as follows. In Section 2, we present the background on bridging and Mandarin definite expressions. In Section 3, we present three experimental studies testing whether demonstratives in Mandarin and English allow uniqueness- and familiarity-based bridging. We show that Mandarin demonstratives are available for bridging, while English demonstratives are significantly degraded. Based on these results, we propose a new analysis for the demonstrative *na* in Section 4. In our analysis, *na* adds a relational predicate and an external relatum argument to the restriction of the definite expression. This analysis makes a prediction that bare nouns would only allow bridging if the noun is inherently relational, which is verified in the final study presented in Section 5. Section 6 concludes with a discussion of our study’s implications.

2 Background: Bridging and Mandarin

2.1 Classic notion of bridging

The term ‘bridging’ can be used broadly to refer to any phenomenon where two chunks of the discourse are linked to each other via some contextual information like coreference due to restrictions such as coherence and relevance (Asher and Lascarides 1998; Hobbs 1979; Matsui 1995; Sperber and Wilson 1986; a.o.).² Clark (1975) uses the term to refer to any mechanism that relates some expression like a definite description, a pronoun, or an epithet to its antecedent. There are many ways in which the intended referent of a definite element can be identified from the context. Ac-

²An anonymous reviewer brings to our attention the observation in Asher and Lascarides 1998 that indefinite expressions support bridging, as demonstrated by the examples in (3).

- (3) a. The table is wobbly. **A leg** needs fixing. [Reviewer]
b. Jack was going to commit suicide. He got **a rope**. [Asher and Lascarides 1998:83]

A proper analysis of the bridging uses of indefinite expressions is beyond the scope of our paper, and we leave this for future investigation. The notion of bridging that we use in this paper is constrained to definite expressions, and this notion crucially depends on the existence presupposition of the definite term and the lack of a suitable antecedent in the context (Clark 1975). This definition applies to the underlyingly partitive case in (3a) but not to the example in (3b).

cording to Clark (1975), the most direct case of bridging is identity with a previously mentioned antecedent as in (4). This is a case of direct anaphora. Note that this does not count as a case of bridging under the sense we are using, but does so in this broader sense.

- (4) I met a man yesterday. {The man, He, The bastard} told me a story. [Clark 1975:170]

More indirect links between the antecedent and the intended referent of a definite expression include what Clark calls indirect reference by association like set membership or necessary parts as in (5) and indirect reference by characterization, as in relevant roles and causes as in (6). The idea is that given some entity or an event, the relevant parts or participants can be easily identified. The antecedent is underlined and the definite expression is bolded.

- (5) a. I met two people yesterday. **The woman** told me a story. [set membership]
b. I looked into the room. **The ceiling** was very high. [necessary parts]
[Clark 1975:171]
- (6) a. I trucked the goods to New York. **The truck** was full. [necessary roles]
b. John went walking at noon. **The park** was beautiful. [optional roles]
[Clark 1975:171–172]

In Clark’s view, the notion of bridging is not confined to definite expressions. For example, reasons that are introduced by *wh*-phrases, as in (7), and concurrences that are introduced by additive particles like *too* and *again*, as in (8), also count as bridging.

- (7) John fell. **What he wanted to do** was scare Mary.
- (8) Alex went to a party last night. He’s going to get drunk **again** tonight.

The bridge linking the contents of the two sentences in (8) is the unsaid assumption that whenever Alex goes to a party, he gets drunk. Clark (1975) notes that bridges do not need to be ‘determinate’, meaning that one could in principle build an ‘infinitely long bridge, or sequence of assumptions, to link one event to the antecedent of the next’ (p. 173). However, he proposes a general pragmatic restriction that one build ‘the shortest bridge possible’ for communication purposes.

In this paper, we focus on the bridging of definite expressions in order to determine which definite mechanisms underlie the use of Mandarin bare nouns and demonstratives. We discuss in the next section the two kinds of bridging we will focus on.

2.2 Two kinds of definiteness and bridging

In English, the antecedent of a regular, non-bridged definite description such as *the linguist* in (9) can be identified through means such as situational uniqueness or familiarity (see Wespel 2008 for a discussion of other definite mechanisms). In the former case, resolution involves identifying a unique linguist in a given situation relevant to the context. In the latter case, resolution involves anaphorically referring to an antecedent introduced in previous discourse.

- (9) I met a linguist. The linguist looked happy.

While English uses *the* for both, Schwarz (2009) shows that many languages morphosyntactically distinguish the two uses. For example, Fering uses two distinct series of articles for the unique and the familiarity uses: the *a*-series for uniqueness and the *di*-series for familiarity (Ebert 1971a). Moreover, German makes a morphophonological distinction, where only the uniqueness-denoting weak form can be contracted when preceded by a preposition. To account for this observation,

Schwarz (2009) proposes two different semantics for the definite article, one for uniqueness and the other for familiarity. The uniqueness-denoting definite in (10a) takes a situation variable s_r and a predicate P and returns the unique x that is P in s_r . The familiarity-denoting definite in (10b) is similar except that it additionally takes a variable y and returns the unique x that is P in s_r and is identical to y . This y serves as the anaphoric index that refers back to the antecedent.

- (10) a. $\llbracket \text{the}_W \rrbracket = \lambda s_r. \lambda P. \exists! x(P(x)(s_r)). \iota x[P(x)(s_r)]$ [Schwarz 2009:81]
 b. $\llbracket \text{the}_S \rrbracket = \lambda s_r. \lambda P. \lambda y. \exists! x(P(x)(s_r) \ \& \ x = y). \iota x[P(x)(s_r) \ \& \ x = y]$ [Schwarz 2009:135]

The unique vs. familiarity distinction in definite expressions is also found in bridging. Schwarz (2009) identifies two types of bridging that correspond to these two meanings: part-whole and producer-product. Part-whole bridging is based on situational uniqueness, where the intended referent of a definite description is identified through a unique existence in the situation established by the context. For example, in (11a), the steering wheel can be identified because there is only one such wheel in the situation that contains the driving event introduced in the first sentence. Producer-product bridging is based on anaphora, where a description that has a relational noun takes an anaphoric index that refers to the antecedent. So in (11b), the relevant author can be identified because the relational noun *author* carries a variable in its denotation that anaphorically refers to the book introduced in the first sentence.

- (11) a. Jane was driving down the street. The steering wheel was cold. [part-whole; Schwarz 2009:11]
 b. Jake bought a book today. The author is French. [relational; Schwarz 2009:11]

While Schwarz (2009) focuses on pairs of nouns that are in a producing relation as in between ‘book’ and ‘author’ in (11b) — hence the name ‘producer-product’ — the crucial notion involved is that of relationality, namely that it is not resolvable merely through situational uniqueness and must rely on relational anaphora to resolve reference. In our study, we broaden the category of ‘producer-product’ to relational noun pairs that a) cannot be resolved through situational uniqueness and b) must rely on relational anaphora. While a more accurate term for this category would be ‘non-containment relational bridging’, we use the term ‘relational bridging’ for short, to refer to this second type of bridging.³

Schwarz (2009) argues that for languages that distinguish between uniqueness-denoting and familiarity-denoting definite markers, the mechanism which underlies bridging is reflected in the grammar. As noted above, in German, the uniqueness-familiarity distinction is reflected in the contraction of the definite article following a preposition. In (12), an example showing a case of part-whole bridging, the contracted form of the definite article is licensed while the full form is degraded, reflecting that the underlying mechanism is uniqueness-based. In (13), where the relevant bridging is that of producer-product, hence relational, the full form of the definite article is licensed while the contracted form is not, suggesting that the underlying mechanism is familiarity-based.

- (12) *Der Kühlschrank war so groß, dass der Kürbis problemlos {im / #in*
 The fridge was so big that the pumpkin without a problem in-the_{weak} / in
dem} Gemüsefach untergebracht werden konnte.
 the_{strong} crisper stowed be could

³As noted by an anonymous reviewer, many of the ‘part-whole’ items we used in our study also involve relational anaphora. This is also the case for part-whole bridging examples in Schwarz 2009. However, the distinction we make is between bridging that can be resolved through situational uniqueness vs. bridging that cannot be. An alternative name for ‘relational bridging’ can be ‘non-unique’ or ‘non-containment’ bridging. We use ‘relational bridging’ for the simplicity of the term. We elaborate on the criteria we used for this category when we discuss the test items.

‘The fridge was so big that the pumpkin could easily be stowed in the crisper.’ [Schwarz 2009:34]

- (13) *Hans entdeckte in der Bibliothek einen Roman über den Hudson. Dabei*
Hans discovered in the library a novel about the Hudson. In the process
fiel ihm ein, dass er vor langer Zeit einmal einen Vortrag {#vom / von
remembered he_{Dat} PART that he a long time ago once a lecture {by-the_{weak} / by
dem} Autor besucht hatte.
the_{strong}} author attended had
‘Hans discovered a novel about the Hudson in the library. In the process, he remembered
that he had attended a lecture by the author a long time ago.’ [Schwarz 2009:129]

For part-whole bridging, the uniqueness-based denotation as in (10a) can apply with some specification on what the relevant situation is. For relational bridging, the familiarity-based denotation in (10b) needs to be modified so that a) it takes a relational noun and b) x is not identified with the index argument but holds some relation to it. The relational version of the strong article proposed in Schwarz 2009 is shown in (14).

- (14) $\llbracket \text{the}_{S; \langle s, \langle \langle e, est \rangle, \langle e, e \rangle \rangle \rangle} \rrbracket = \lambda s_r. \lambda R. \lambda z. \lambda x [R(y)(x)(s_r) \ \& \ y = z]$ [Schwarz 2009:141]

To summarize, in bridging, situational uniqueness and relational anaphora can be teased apart through context manipulation. Thus, we can test different definite expressions across languages in respective contexts to determine whether uniqueness and/or familiarity play a role in licensing the expression. In this study we investigate how native speakers accept and produce sentences with bare nouns and demonstratives in the two bridging contexts to investigate how these expressions divide up the definite space. Before we move on to our studies, we first present a background on Mandarin bare nouns and demonstratives, including some previous accounts of them.

2.3 Mandarin demonstratives and bridging

Mandarin is a language that does not have an obvious counterpart to the English definite determiner. Instead, bare nouns, as in (15a), which occur freely in the language, as well as demonstrative descriptions containing the demonstrative *na* and a classifier followed by the noun, as in (15b), allow definite readings. We call these expressions bare nouns and demonstrative descriptions, respectively, throughout the paper.⁴

- (15) a. *gou yao guo malu.*
dog want cross road
‘The dog wants to cross the road.’
b. *na tiao gou yao guo malu.*
that CL dog want cross road
‘That dog wants to cross the road.’

Semantic analyses of definite bare nouns and demonstrative descriptions in Mandarin vary in the literature. In the rest of this section, we review two recent accounts of these definite expressions. The first account we discuss is from Jenks 2018 and the second is from Dayal and Jiang 2021.

⁴In the linguistic examples throughout this paper, we translate bare nouns as English definites with ‘the’ to reflect their definite reading. Mandarin bare nouns also have indefinite, generic, and kind uses, and these readings may be constrained by contexts. Moreover, we translate *na* as the English demonstrative *that*, without committing to the theoretical claim that they should be analyzed as semantic equivalents.

Jenks (2018) argues that the difference between the bare noun and the demonstrative description containing *na* aligns with the uniqueness vs. familiarity distinction made in Schwarz 2009, where uniqueness-based definiteness is expressed with bare nouns and familiarity-based definiteness is expressed with demonstratives. He supports this argument by showing that bare nouns are licensed in uniqueness-based contexts such as (16) and (17). The difference between (16) and (17) is the size of the domain in which the unique entity is identified. In (16), *the moon* is licensed because based on our common knowledge there is a unique moon for our planet. The bare noun *yueliang* ‘moon’ refers to that unique moon. In (17), the domain is smaller and refers to a context where some individual finished some specific bowl of soup. The bare noun *tang* ‘soup’ refers to the unique soup that the individual finished. These align with those that have been shown to license unique definites in other languages (Schwarz 2009, 2013).

- (16) *yueliang sheng shang lai le.*
 moon rise up come ASP
 ‘The moon has risen.’ [Jenks 2018:507; originally Chen 2004:1165]
- (17) *Hufei he wan le tang.*
 Hufei drink finish ASP soup
 ‘Hufei finished the soup.’ [Jenks 2018:504; originally Cheng and Sybesma 1999:510]

Especially relevant to our paper is Jenks’ observation that part-whole bridging is expressed by bare nouns, as in (18).

- (18) *chezi bei jingcha lanjie le yinwei mei you tiezhi zai paizhao shang.*
 car PASS police intercept ASP because NEG have sticker at license.plate on
 ‘The car was intercepted by the police because there wasn’t a sticker on the license plate.’
 [Jenks 2018:508]

Jenks notes that anaphoric uses of bare nouns are much more restricted. For example, in an anaphoric context as in (19), a bare noun is infelicitous and a demonstrative is needed instead.

- (19) *jiaoshi li zuo-zhe yi ge nansheng he yi ge nusheng. wo zuotian yudao*
 classroom inside sit-PROG one CL boy and one CL girl I yesterday meet
 #(*na ge*) *nansheng.*
 that CL boy
 ‘There are a boy and a girl sitting in the classroom. I met the boy yesterday.’
 [Jenks 2018:510]

In order to account for these observations, Jenks (2018) proposes the following denotations. A definite interpretation of bare nouns involves a typeshifter presented in (20a), which takes the NP predicate and returns the unique entity in some given situation that meets the denotation of the predicate. The demonstrative *na* projects a syntactic DP and realizes (20b), which has one further argument slot for domain restriction. In addition to the NP predicate *P* and the resource situation *s_r*, the $\langle e, t \rangle$ predicate *Q* serves to restrict the domain in which uniqueness is evaluated. The domain restriction *Q* differs from the index in Schwarz 2009 in that a) it can be extended to any individual-denoting element like names or pronouns and compose with the rest of the semantics through PRED (Partee 1987) which turns it into an $\langle e, t \rangle$ predicate; and b) it is in the presuppositional domain.

- (20) a. $\llbracket t \rrbracket = \lambda s_r. \lambda P_{\langle e, st \rangle} : \exists! x [P(x)(s_r)]. \iota x [P(x)(s_r)]$
 b. $\llbracket t^x \rrbracket = \lambda s_r. \lambda P_{\langle e, st \rangle}. \lambda Q_{\langle e, t \rangle} : \exists! x [P(x)(s_r) \wedge Q(x)]. \iota x [P(x)(s_r)]$

Jenks (2018) further proposes a principle that maximizes the use of index whenever possible (*Index!*), explaining why bare nouns are ruled out and demonstratives are required in anaphoric contexts like (19), even when many anaphoric contexts entail uniqueness as well. Finally, he notes that this generalization has an exception: in subject positions, bare nouns can be anaphoric due to their topic status.

However, that bare nouns are restricted to unique contexts through *Index!* has been contested in the literature. There are many examples in Mandarin where (subject and non-subject) bare nouns are licensed and resolved to antecedents present in previous discourse such as intersentential anaphora (Ahn 2019; Bremmers et al. 2022; Dayal and Jiang 2021; Jiang 2017; a.o.). For example, Bremmers et al. (2022) show through a corpus study that Mandarin bare nouns are quite readily licensed in both unique and anaphoric contexts, which is not predicted by *Index!*.

Dayal and Jiang (2021) also present arguments against the empirical generalization in Jenks 2018, arguing that bare nouns can be used in anaphoric contexts. They propose that Mandarin bare nouns be analyzed on par with English *the* and that Mandarin *na* be analyzed on par with English *that*. They argue that, as semantic equivalents, Mandarin *na* and English *that* carry an anti-uniqueness presupposition. Specifically, demonstratives with the description *N* require that there be another entity that meets the *N* description outside the minimal situation in which the main predicate is evaluated. Unless there is a reason to ‘extend’ the minimal situation to include other entities, demonstratives are predicted to be degraded, in English and Mandarin alike. This would explain why in (21) the English demonstrative description is degraded while the definite is fine: the continuation does not introduce a new entity, namely a roof that is not the roof of the mentioned house, so there is no reason to extend the situation and accommodate another roof in the process.

- (21) Mary bought a house.
- a. The roof needed to be replaced.
 - b. #That roof needed to be replaced.

Given that there is a debate on how different kinds of mechanisms licensing definiteness such as uniqueness and familiarity map onto Mandarin definite expressions, we apply the strategy outlined above to test whether bare nouns and demonstratives license different kinds of bridging. The two proposals reviewed above make different predictions on the availability of bridging uses of demonstrative descriptions and definite bare nouns. Jenks (2018) predicts bare nouns to allow part-whole bridging only and demonstratives to allow relational bridging only. Dayal and Jiang (2021) predict bare nouns to allow both types of bridging and demonstratives to allow neither, patterning with English *that*.

In Section 3, we present experimental studies that explore whether bare nouns and demonstratives allow part-whole (uniqueness-based) and relational (familiarity-based) bridging in Mandarin and English. Using an off-line ratings task (Study 1) and a production study (Study 3), we show that in Mandarin both kinds of bridging are possible with bare nouns and demonstratives. We compare the results of Study 1 to English (Study 2) and show that the two languages are clearly different in the availability of demonstratives in bridging contexts.

3 Bridging in Mandarin and English: Experimental studies

In this section, we present three studies that explored bridging in Mandarin and English. In Sections 3.2 and 3.3, we discuss off-line ratings tasks we conducted in Mandarin (Study 1) and English (Study 2), investigating whether definite expressions in the two languages license part-whole and relational

bridging. In Section 3.4, we present a production study on Mandarin (Study 3).

Before presenting the studies, we first discuss the criteria we used to create part-whole and relational bridging contexts for our stimuli. Because cross-linguistic variation is inevitable, we try to describe our categorization of bridging as clearly as possible for potential adaptations to other languages.

3.1 Identifying different types of bridging in Mandarin

We discuss in this section how we categorized part-whole and relational bridging in Mandarin for our experimental stimuli. The general scheme we used in all of our studies contained two clauses, where the first created the relevant context for bridging and the second contained a bridged definite expression, either a bare noun or a demonstrative description. From here on, we will use the term ‘antecedent’ to refer to either the first sentence (‘antecedent sentence’) or an entity that stands in a one-to-one relation to the bridged noun (‘antecedent noun’). The bridged definite expression will be called ‘the bridged noun’ or ‘the definite expression’.

3.1.1 Part-whole bridging

For part-whole bridging, we created contexts such that there is a unique entity that meets the denotation of the bridged noun in the situation established by the previous context (Schwarz 2009). For example, in (22), the first clause establishes a context where there is a bike in the backyard. The minimal situation created contains a bike and a backyard where the bike is located. The second clause then refers to a seat. Although a seat is not previously introduced, it can be identified within the minimal situation established in the first clause because, under general assumptions, there is a unique seat to a bike.

- (22) *zixingche zai houyuan li, wo zhunbei qu ca yixia na ge chezuo.*
 bike at backyard inside I plan go wipe once that CL seat
 ‘The bike was in the backyard. I planned to wipe that seat.’

Other examples of part-whole bridging stimuli included pairs such as house-roof, fridge-crisper, and pen-cap.

3.1.2 Relational bridging

The second type of bridging discussed in Schwarz 2009 is called ‘producer-product’ and crucially involves a producer-product relation between the antecedent noun in the first sentence and the bridged expression in the second: the entity denoted by the antecedent noun (e.g. ‘a book’) is ‘produced’ by the entity denoted by the definite expression (e.g. ‘the author’). As noted in Schwarz 2009, the crucial component of the producer-product relation is that of a relational reference between the bridged noun and the antecedent. Namely, the bridged noun has to be relational, carrying a variable in its semantic denotation that refers to the antecedent. Note, however, that relationality is not exclusive to producer-product bridging. Many examples of part-whole bridging can also be described in terms of relational anaphora. The distinction between part-whole and producer-product then boils down to whether situational uniqueness can resolve the reference or not: in the former, situational uniqueness would do, while in the latter, the unique existence of the antecedent is not entailed in the minimal topic situation provided by the previous sentence, and thus reference relies on additional mechanism such as relational anaphora.⁵ Thus, in forming our relational bridging items,

⁵We thank an anonymous reviewer for helping us clarify this point.

we broadened the concept of producer-product to all relational links and ensured that resolution through situational uniqueness is not possible.

We created contexts where there is a one-to-one relation between the bridged noun and the antecedent, with the bridged noun being relational, in other words requiring the presence of the antecedent conceptually. In order to exclude the possibility of the situational uniqueness mechanism being available in this group of stimuli, we made sure that the referent of the bridged noun was not physically contained inside the situation established by the antecedent sentence. For example, in (23), the author cannot be identified inside the minimal situation that contains the event of buying a book.

- (23) *zuotian wo mai le **shu**. wo hen xiang jianjian na wei **zuozhe**.*
 yesterday I buy ASP book I very want meet that CL author
 ‘Yesterday I bought the book. I really want to meet that author.’

Other examples of relational bridging included pairs such as painting-painter, movie-director, lock-key, phone-charger, and seat-ticket. We considered terms like a key, a password, a remote, and a charger to be relational. Note that while relational nouns are present across languages, it is hard to define them categorically. In general, relational nouns are assumed to take two arguments: one for the entity that meets the denotation of the description and another for an entity that holds a certain relation with the former entity (Barker 2011; cf. Asudeh 2005; Meyers 2007). There are several criteria for distinguishing relational nouns from regular sortal nouns (Barker 2011; Löbner 1985; a.o.). Unlike sortal nouns, relational nouns in English can take an additional overt argument introduced with the preposition *of*. Compare the relational noun *child* vs. the sortal noun *person*:

- (24) a. child of someone
 b. person (*of someone)

Semantically, relational nouns are always defined by one (or several) specific entities that they are related to. According to Barker 2011, ‘a day counts as a birthday only in virtue of standing in a certain relationship to a person’ (p. 3). Thus, the inherent definition of the relational noun requires the related entity.

We can find a parallel diagnostic for Mandarin relational nouns with the possessive *de*, as in (25). While the relational noun *haizi* ‘child’ can take the additional possessor introduced with *de*, the sortal noun *ren* ‘person’ cannot.

- (25) a. *mou-ren de haizi*
 some-person DE child
 ‘child of someone’
 b. **mou-ren de ren*
 some-person DE person
 Intended: ‘person of someone’

However, this diagnostic is not as reliable for Mandarin as for English. As shown in (26), the sortal noun *hua* ‘flower’ can enter this construction as well.

- (26) *mou-ren de hua*
 some-person DE flower
 ‘flower of someone’

Thus, in identifying relational nouns, we relied on the syntactic criterion of allowing a genitive argument with *de* as well as semantic cues. For example, if a noun failed the *de*-test, we categorized

it as a sortal noun. If a noun passed the *de*-test but does not have a conceptual requirement of another entity for the meaning to be complete (e.g. flower), then we did not use it in our relational stimuli. For example, a key passes the *de*-test (*suo de yaoshi* ‘key of a lock’) and semantically is defined in virtue of standing in some relation to a lock. A password also passes the *de*-test (*zhanghu de mima* ‘password of an account’) and is defined in virtue of standing in some relation to an account. Thus, we assume that the bridged expressions we used in our stimuli are relational and parallel to the producer nouns of producer-product as defined in Schwarz 2009.

It is not possible to concretely argue that these stimuli always triggered part-whole or relational bridging when the participants were processing them. As Clark (1975) notes, it is possible to build an infinitely long and complex bridge between the antecedent and the anaphor. However, given that participants are more likely to build the most salient kind of bridge available, we assume that our part-whole and relational items represented uniqueness- and familiarity-based bridging, respectively.

Given these definitions for part-whole and relational bridging, we explored how Mandarin and English compare in licensing the two kinds of bridging with their definite expressions using off-line naturalness ratings. Study 1 compares Mandarin bare nouns and demonstratives, while Study 2 compares English definite and demonstrative descriptions. We discuss each study in detail below.

3.2 Study 1: Mandarin Sentence Ratings Task

Study 1 was a Sentence Ratings task where participants were presented with sentences and asked to rate how natural they were. This experiment served as the baseline for our study, because there has not been any previous work that formally tested the difference between bare nouns and demonstratives in part-whole and relational bridging. The goal was to see how bridging is rated by native speakers of Mandarin, and to investigate the effect of bridged noun types on the ratings. Thus, we manipulated two main factors: BRIDGING TYPE (part-whole vs. relational) and BRIDGED NOUN TYPE (bare noun vs. demonstrative).⁶

3.2.1 Methodology

Our target stimuli contained 20 bridging sentence pairs. We manipulated the bridging type, which varied between part-whole and relational bridging, following the criteria discussed in Section 3.1. The English translations of the stimuli are presented in (27).

- (27) a. **Part-whole bridging items** (antecedent noun, bridged noun):
 (car, steering wheel), (computer, screen), (kettle, handle), (church, tower), (fridge, crisper), (bike, seat), (pen, cap), (house, roof), (safe, door), (glass, bottom)
- b. **Relational bridging items** (antecedent noun, bridged noun):
 (novel, author), (painting, painter), (movie, director), (CD, singer), (stand-up show,

⁶In a previous version of this study, we included three additional factors: ANTECEDENT NOUN TYPE, ANIMACY, and SYNTACTIC POSITION. First, we manipulated the noun type of antecedents (an indefinite description, a bare noun, or a demonstrative), based on the experimental findings in Law and Syrett (2017), which suggest that the syntactic form of the antecedent can influence subsequent anaphora. Second, we varied the animacy of both the antecedents and the bridged nouns in our stimuli, given a heavy empirical focus on inanimate part-whole bridging and animate producer-product bridging in previous literature. We explored different combinations of animacy and bridging type to investigate whether animacy plays a role in facilitating bridging. Third, we manipulated the syntactic positions of the antecedent and bridged nouns (subject position vs. object position in a simple declarative sentence). Jenks (2018) predicts that bare nouns are sensitive to syntactic positions and support an anaphoric use in subject positions due to their topic status, so we checked whether syntactic position affected the results significantly.

The results from the previous study revealed no significant main effect of these factors. Therefore, we did not include these factors in this new study and simplified our models, following the suggestion of the anonymous reviewers.

comedian), (door lock, key), (bank card number, password), (TV, remote control), (phone, charger), (seat, ticket)

We further manipulated the type of the bridged noun. The noun appeared as either a bare noun, such as (28a), or a demonstrative construction with *na* and a classifier, such as (28b).

- (28) a. *che*
car
'car' (BN)
- b. *na liang che*
that CL car
'that car' (DEM)

We controlled for the antecedent noun type and the syntactic position of both antecedent and bridged nouns. First, all antecedent nouns in the target items were bare nouns. Mandarin bare nouns can have either a definite or an indefinite interpretation, and we left it for the participants to choose the most natural interpretation. Second, both the antecedent and bridged nouns appeared in the object position — either as the direct object of the matrix verb, or following a preposition. The full set of stimuli items is provided in the Supplementary Materials.

We present two target stimuli as examples in (29). In (29a), the part-whole bridging item contains a bare noun antecedent *zixingche* 'bike' and a demonstrative construction as the bridged noun *na ge chezuo* 'that seat'. The seat, as a part of the bike, can be uniquely identified. In (29b), the relational bridging item contains a bare noun antecedent *changpian* 'CD' and a bare bridged noun *geshou* 'singer'. There is a one-to-one association between the two nouns and the singer is not physically contained in the CD.

- (29) a. *Chen Haoran ba zixingche ting zai le houyuan, jieguo ta de gou*
Chen Haoran BA bike park at ASP backyard as.a.result he POSS dog
zhua-po le na ge chezuo.
scratch-break ASP that CL seat
'Chen Haoran parked his bike in the backyard, and then his dog scratched that seat.'
- b. *Chen Yue shang-zhou mai-lai le changpian, ta mei-tian dou xiang-zhe qu*
Chen Yue last-week buy-come ASP CD she every-day DOU think-PROG go
zhao geshou qian-ming.
find singer sign-name
'Chen Yue bought a CD last week, and every day she dreams about obtaining an autograph from the singer.'

The 20 sets of target stimuli yielded a total of 40 target items. We adopted a within-subject design: each participant saw both part-whole and relational bridging sentences in the study. We also ensured that each participant only saw one variation within each stimuli set, resulting in a total of 20 target sentences per participant.

In addition to the target stimuli, we included 30 syntactically well-formed controls that were either semantically odd, pragmatically odd, or neutral. Participants were asked to rate how natural the given sentences are. We discuss the motivations for the particular instruction language we used and the control stimuli below.

Instruction language It is known that the methods used in an experiment such as the instruction and the response scale have a considerable impact on the results (Kirk 2012; Myers 2017; Schütze

2005; Sprouse et al. 2013). This is especially the case for studies looking at semantic and pragmatic oddness, since the ratings often measure subtle differences. In order to maximize the information we get from the rating responses and to ensure consistency in the measurement, we designed our ratings study based on Zhu and Ahn 2023. In Zhu and Ahn 2023, it was shown that the terms *natural* in English and *ziran* ‘natural’ in Mandarin in the instruction maximize the rating difference between semantically and pragmatically odd sentences in both English and Mandarin, though the two languages show differences in the range of responses, highlighting the importance of language-specific and study-internal norming procedure.

Thus in our ratings task, we presented participants with sentences involving different kinds of bridging and definite expressions and asked them to rate how ‘natural’ the sentences are. The instruction we used is shown in (30).

- (30) *qing gei juzi de ziran chengdu dafen. 1 fen wei zui bu ziran, 7 fen wei zui ziran.*
 please give sentence DE natural degree rate 1 point be most not natural 7 point be
 most natural
 ‘Please rate the naturalness of the sentence(s). 1 means least natural, and 7 means most natural.’

Control stimuli As discussed above, different languages show different ranges of responses, meaning that it is impossible to directly compare the rating responses between two languages. To form a standard for comparison, we added control stimuli as filler items, following Zhu and Ahn 2023: 10 sentences that were semantically odd, 10 sentences that were pragmatically odd, and 10 sentences that were neutral. Semantically odd sentences included logical contradictions or thematic mismatch, as shown in (31a). Pragmatically odd sentences included repetitions of information as in (31b). Neutral sentences did not have any obvious linguistic violations, as shown in (31c).

- (31) a. *Zhang Xiaoming shi ge yihun-de danshenhan, wo he ta hen shu.*
 Zhang Xiaoming be CL married bachelor I and he very close
 ‘Zhang Xiaoming is a married bachelor. He and I are very close.’ [semantically odd]
 b. *zuotian xiayu de shihou xiayu le.*
 yesterday rain DE time rain ASP
 ‘Yesterday it was raining when it was raining.’ [pragmatically odd]
 c. *Zhou Anyi zhengli hao keben, jue ding jintian qu-shangxue.*
 Zhou Anyi organize complete textbook decide today go.to.school.
 ‘Zhou Anyi organized the textbooks and decided to go to school today.’ [neutral]

By comparing the ratings of the target sentences against these three types of control stimuli, we can more precisely pinpoint where the target sentences lie in terms of naturalness against a scale that covers different kinds of linguistic violations.

We recruited 173 native Mandarin speakers (18–64; gender-balanced) via Prolific. Participants were redirected to a Prolific survey, where they were asked to first provide some demographic and language background information and then complete the sentence ratings task. Participants were paid \$2–3 for their time.

Each participant was presented with a total of 50 stimuli, randomized in order: 10 part-whole and 10 relational bridging sentences (pseudo-randomized in referent noun type, animacy, and syntactic position), and 30 controls. Participants were asked to rate the naturalness of these sentences on a 7-point Likert scale, as in Fig. 1.

陈浩然把自行车停在了后院，结果他的狗抓破了那个车座。

请给句子的自然程度打分，1分为最不自然，7分为最自然。

最不自然 ○ 1 ○ 2 ○ 3 ○ 4 ○ 5 ○ 6 ○ 7 最自然

Figure 1: Sample question (a target item [DEM anaphor, part-whole]) in Study 1 (Mandarin)

3.2.2 Results

We excluded data from 13 participants who rated the neutral controls below 5. In this section, we report the results based on data from 160 participants. To compare ratings across conditions, we fit a Cumulative Link Mixed Model (CLMM) using the ordinal package (Christensen 2022; Bross 2019) in R. The best-fitting model included BRIDGING TYPE and BRIDGED NOUN TYPE as fixed effects to assess their influence on ratings. To account for variability across participants and items, we included PARTICIPANT and ITEM as random effects, incorporating both random intercepts and random slopes for NOUN TYPE. By including random slopes, we allowed for potential variation in the effect of noun type across participants and items, thereby improving model fit. Model selection was based on AIC to ensure the best-fitting model for our data.

The modeling results showed a main effect of **bridged noun type** ($\beta = -0.29$, $SE = 0.11$, $z = -2.59$, $p = 0.009$) and a marginal effect of **bridging type** ($\beta = 0.39$, $SE = 0.23$, $z = 1.65$, $p = 0.099$). No interaction was found between the two factors. The ratings of all bridging sentences were significantly higher than those of pragmatically odd ($\beta = -4.38$, $SE = 0.09$, $z = -50.52$, $p < 0.001$) and semantically odd sentences ($\beta = -5.03$, $SE = 0.10$, $z = -49.69$, $p < 0.001$).

Fig. 2 plots the average ratings for part-whole and relational bridging, color-coded by bridged noun type. The figure also shows the average ratings for the control sentences: semantically odd sentences (red dashed line, average rating: 1.62), pragmatically odd sentences (orange dashed line, average rating: 1.99), and neutral sentences (black dashed line, average rating: 6.06). On average, part-whole bridging sentences were rated 5.49 (5.59 for bare nouns as bridged nouns and 5.40 for demonstratives as bridged nouns), and relational bridging sentences were rated 5.70 (5.80 for bare nouns as bridged nouns and 5.61 for demonstratives as bridged nouns).

To further examine item-level effects, we fit a Cumulative Link Model (CLM) via the ordinal package. The model included NOUN TYPE as a fixed effect and no random effect. Among the 20 bridging sentence sets, only 4 sets showed a significant effect of bridged noun type on ratings. The noun pairs in these items are listed in (32), along with their p -values, noun type preferences, and bridging types.

- (32) a. PW1, (car, steering wheel): $p = 0.00134$, BN > DEM; part-whole bridging
b. REL9, (phone, charger): $p = 0.03$, BN > DEM; relational bridging

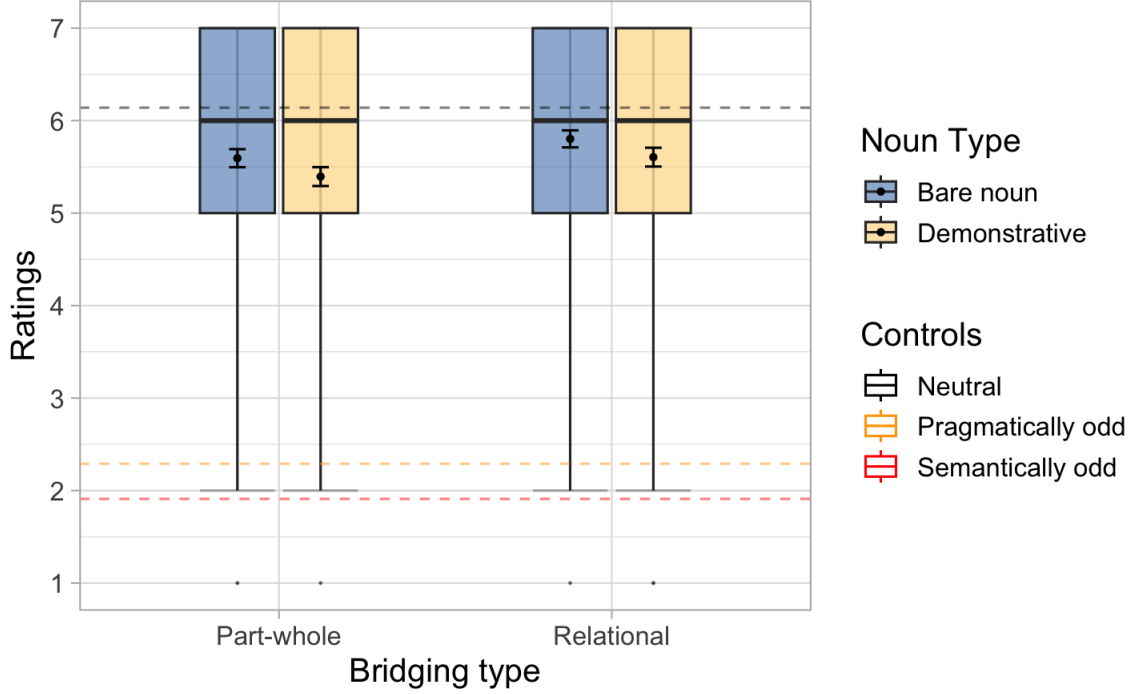


Figure 2: Study 1: Ratings as function of bridging types, color-coded by bridged noun type (Mandarin)

- c. REL10, (seat, ticket): $p = 0.000964$, BN > DEM; relational bridging
- d. REL2, (painting, painter): $p = 0.02$, DEM > BN; relational bridging

To show the item variation, Fig. 3 plots the average ratings for each item, color-coded by bridged noun type, and Fig. 4 plots the rating distribution for each item, also color-coded by bridged noun type. In the next section, we discuss possible explanations for this variation.

3.2.3 Discussion

Our results indicate that both bare nouns and demonstratives are natural in part-whole and relational bridging. However, demonstratives were systematically rated lower than bare nouns across both bridging types. At the item level, the difference in rating between bare nouns and demonstratives was not significant for most items, with four exceptions listed in (32). We discuss these cases below.

The relational bridging pair (painting, painter) was the only instance where demonstratives were rated significantly higher than bare nouns. We attribute this to the presence of the stative verb *renshi* ‘know’ in the continuation, as shown in (33). This verb in combination with the bare noun *huaqia* ‘painter’ might have facilitated a generic interpretation, suggesting that Zhou doesn’t know any individual who belongs to the category of painters. Since Mandarin demonstrative descriptions do not allow such a generic reading, the demonstrative condition aligns more naturally with the preceding context than the bare noun condition. Thus this seems to be caused by an idiosyncratic property of the predicate ‘know’ and its interaction with the bare noun.

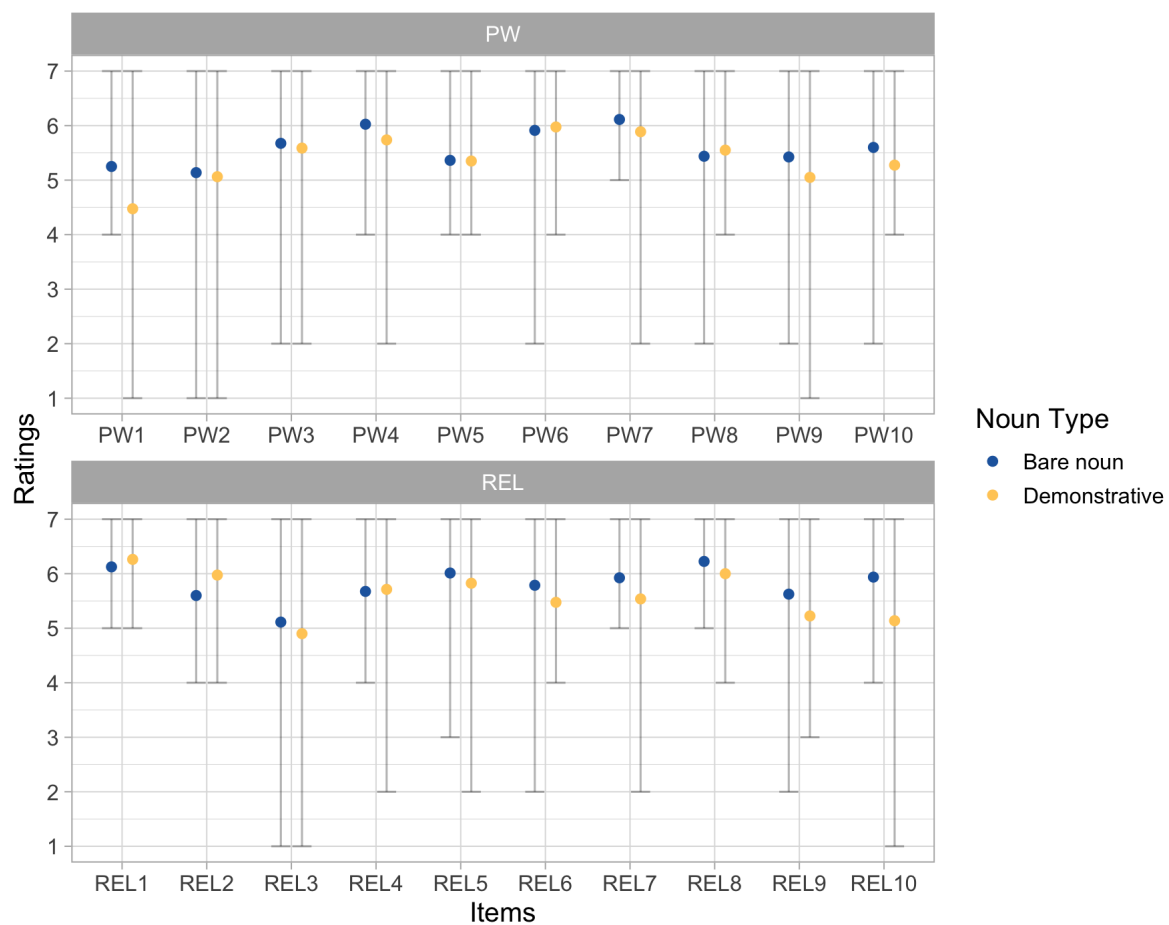


Figure 3: Study 1: Ratings as function of items, color-coded by bridged noun type (Mandarin)

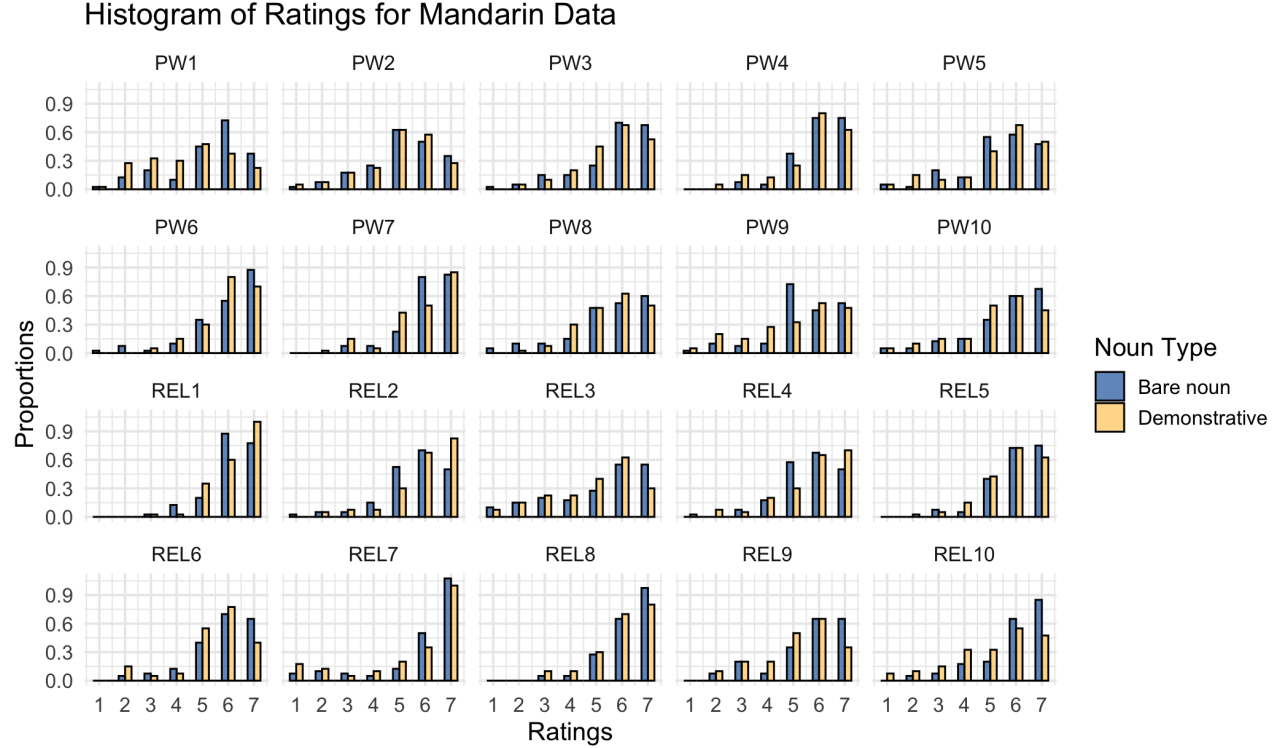


Figure 4: Study 1: Histograms of ratings by items, color-coded by bridged noun type (Mandarin)

- (33) *Zhou Yuxuan jia-li gua-zhe youhua, dan ta wanquan bu renshi (na ge) huajia.*
 Zhou Yuxuan home-in hang-PROG painting but he totally NEG know DEM CL painter
 ‘Zhou Yuxuan has a painting hanging at home, but he doesn’t know the/that painter.’

For the other items with significant rating differences, we suspect that the naturalness of the demonstrative condition is affected by additional linguistic factors: an infrequent combination of *fangxiangpan* ‘steering wheel’ with the classifier *ge* in (34a), and a weak uniqueness association between a phone and a charger in (34b) and between a seat and a ticket in (34c).

- (34) a. *Lin Zixuan xin mai-le qiche, dan ta hen-kuai faxian you-ren touzou-le*
 Lin Zixuan new buy-ASP car but he very-fast discover have-person steal.away-ASP
(na ge) fangxiangpan.
 DEM CL steering.wheel
 ‘Lin Zixuan bought a car, but he quickly discovered that someone had stolen the/that steering wheel.’
- b. *Liu Jiayi dao bangongshi shi zheng zhunbei yong shouji, ta zhe cai*
 Liu Jiayi arrive office when PROG prepare use phone she this then
yishi-dao ziji mei dai (na kuai) chongdianqi.
 realize-arrive self NEG bring DEM CL charger
 ‘Liu Jiayi was planning to use her phone when she arrived at the office, and then she realized that she forgot to bring the/that charger.’

- c. *Chen Junyu shang che hou zhao-dao le zuowei, danshi ta wangji dai*
 Chen Junyu get.on train after look.for-arrive ASP seat but he forget bring
(na zhang) chepiao le.
 DEM CL ticket ASP
 ‘Chen Junyu got on the train and found his seat, but he forgot to bring the/that ticket.’

Overall, the majority of target items in Study 1 did not show significant differences in ratings between bare nouns and demonstratives, regardless of bridging type. We take this to suggest that native speakers readily accept both noun forms in bridging contexts. To further examine the effect size of bridged noun types and assess the naturalness of demonstratives in Mandarin, we present an English version of this study in Section 3.3 and compare the rating patterns across languages in Section 3.3.4.

3.3 Study 2: English Sentence Ratings Task

We conducted an English sentence rating task (translated from Study 1) to see if the English demonstrative *that* patterns similarly to Mandarin *na*. Based on ratings from 166 native English speakers, we found that demonstrative constructions were rated significantly lower than definite descriptions, showing a larger effect compared to Mandarin (see Section 3.3.4 for a statistical comparison). These experimental results confirm the commonly adopted assumption that demonstratives are dispreferred in bridging contexts — both part-whole and relational — although the judgments are not entirely categorical. In this section, we report this study in detail.

3.3.1 Methodology

The target stimuli for this study were English translations of stimuli used in the Mandarin sentence ratings task (Study 1). We included 10 part-whole and 10 relational bridging sentence pairs, systematically manipulating the bridged noun type (definite vs. demonstrative). The full list of bridging pairs is presented in (27).

We show examples of two target stimuli in (35): (35a) is a part-whole bridging item where the bridged noun is a definite description, *the seat*. Example (35b) illustrates a relational bridging item, where the bridged noun is a demonstrative description, *that singer*.

- (35) a. David parked **his bike** in the backyard, and then his dog scratched **the seat**.
 b. Grace bought **a CD** last week, and every day she dreams about obtaining an autograph from **that singer**.

Antecedent nouns were assigned syntactic forms that allowed for the most natural interpretation. These forms included possessive determiners (e.g. *her*), indefinite articles (*a*), and definite articles (*the*). The antecedent nouns appeared as objects of either the matrix verb or a preposition.

As in Study 1, we included 30 syntactically well-formed controls: 10 sentences with semantic oddness, 10 sentences with pragmatic oddness, and 10 sentences that were neutral.

We recruited 177 native English speakers (18–64; gender-balanced) via Prolific. Participants were redirected to a PCIbex survey, where they were asked to first provide demographic and language background information and then complete the sentence ratings task. Participants were paid \$2–3 for their time. Each participant was presented with a total of 50 stimuli, randomized in order: 20 bridging sentences and 30 controls. Participants were asked to rate the naturalness of these sentences on a 7-point Likert scale, as in Fig. 5.

Grace bought a CD last week, and every day she dreams about obtaining an autograph from the singer.

Please rate the naturalness of the sentence (1 = least natural, 7 = most natural).

Least natural ○ ○ ○ ○ ○ ○ ○ Most natural
1 2 3 4 5 6 7

Figure 5: Sample question (a test item [DEF anaphor, relational]) in Study 2 (English)

3.3.2 Results

We excluded data from 11 participants whose ratings for the neutral control sentences were below 5. The results reported in this section are based on data from 166 participants. To compare ratings across conditions, we fit a Cumulative Link Mixed Model using the ordinal package in R (Christensen 2022; Bross 2019). The best-fitting model included BRIDGING TYPE and BRIDGED NOUN TYPE as fixed effects, and PARTICIPANT and ITEM as random effects, with both random intercepts and random slopes for NOUN TYPE. Model selection was based on AIC.

The modeling results showed a main effect of **bridged noun type** ($\beta = -1.44$, $SE = 0.14$, $z = -9.93$, $p < 2e - 16$) and **bridging type** ($\beta = 0.85$, $SE = 0.23$, $z = 3.66$, $p = 0.000254$). No interaction between the two factors was found. Ratings of all bridging sentences were significantly higher than those for pragmatically odd ($\beta = -3.07$, $SE = 0.07$, $z = -44.39$, $p < 2e - 16$) and semantically odd sentences ($\beta = -3.67$, $SE = 0.08$, $z = -47.28$, $p < 2e - 16$).

Fig. 6 plots the average ratings for part-whole and relational bridging, color-coded by bridged noun type. The figure also includes the average ratings for the control sentences: semantically odd sentences (red dashed line, average rating: 1.91), pragmatically odd sentences (orange dashed line, average rating: 2.29), and neutral sentences (black dashed line, average rating: 6.14). On average, part-whole bridging sentences were rated 4.83 (5.35 for definite descriptions as bridged nouns and 4.32 for demonstratives as bridged nouns), and relational bridging sentences were rated 5.46 (5.91 for definite descriptions as bridged nouns and 5.01 for demonstratives as bridged nouns).

To compare ratings across conditions by item, we fit a Cumulative Link Model via the ordinal package. The model included NOUN TYPE as a fixed effect and no random effects. Among the 20 bridging sentence sets, only 2 sets did **not** show a significant effect of bridged noun type on ratings. For all other sets, ratings for definite descriptions were significantly higher than for demonstratives. The noun pairs used for these two items are listed in (36), along with the p -values from the CLM results and their bridging types.

- (36) a. REL1, (novel, author): $p = 0.189$; relational bridging
b. REL3, (movie, director): $p = 0.124$; relational bridging

To illustrate the variation by item, Fig. 7 plots the average ratings for each item, color-coded by bridged noun type. Fig. 8 plots the rating distribution for each item, also color-coded by bridged

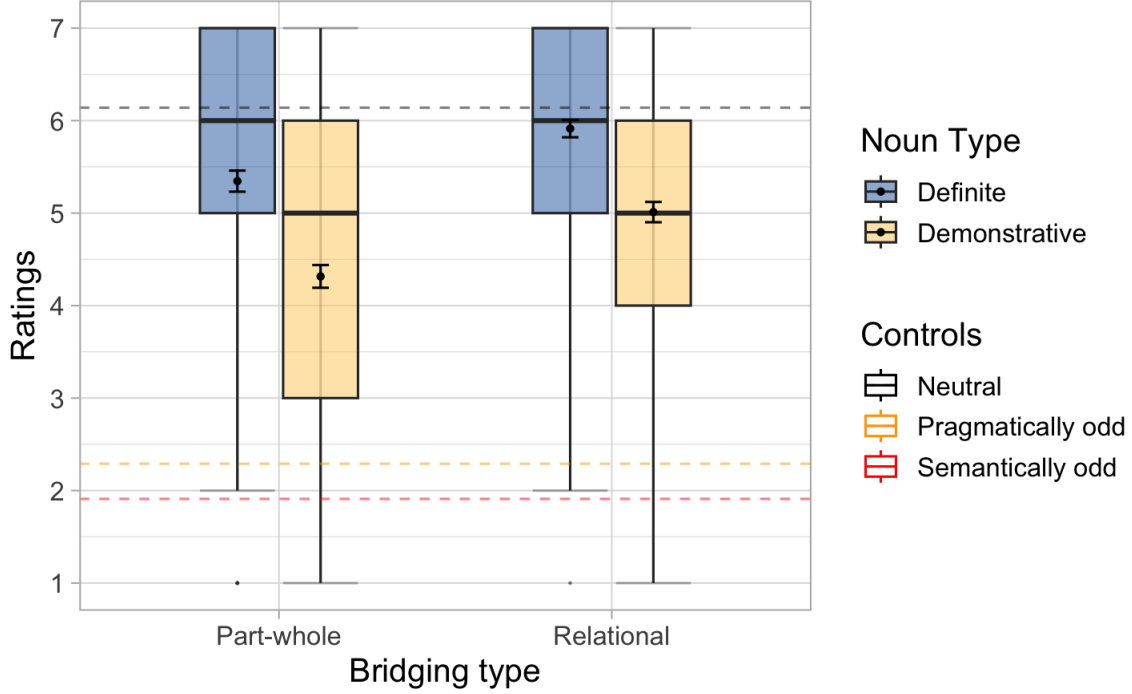


Figure 6: Study 2: Ratings as function of bridging types, color-coded by bridged noun type (English)

noun type. We discuss possible reasons for the item variation in the following section.

3.3.3 Discussion

Our results show that English demonstrative descriptions (with *that*) are generally dispreferred in bridging sentences, in comparison to English definite descriptions. Specifically, the bridged noun type had a significant effect, with demonstrative constructions being systematically rated lower than definite descriptions, regardless of the bridging type. This finding aligns with the common assumption that demonstratives disallow bridging (Dayal and Jiang 2021; Himmelmann 1996).

While the significance of bridged noun type was observed in both English and Mandarin, the by-item analysis reveals a cross-linguistic difference: the effect size of bridged noun type differs between the two languages. This distinction is to be statistically confirmed in Section 3.3.4. The by-item modeling results show that while the bridged noun type did not significantly affect the ratings for most of the Mandarin bridging sentences (with only 4 out of 20 sentences showing significance), it did result in significant differences for the majority of the target items in the English study (with 18 out of 20 showing significance).

The by-item analysis highlights two target items that failed to show statistically significant differences between definite and demonstrative descriptions as the bridged noun. Both items involve relational bridging, and they are presented in (37).

- (37) a. Lucas read a novel yesterday, and if possible, he really wants to meet the/that author.
b. Ryan watched a movie last week, and he believed that he ran into the/that director after the movie.

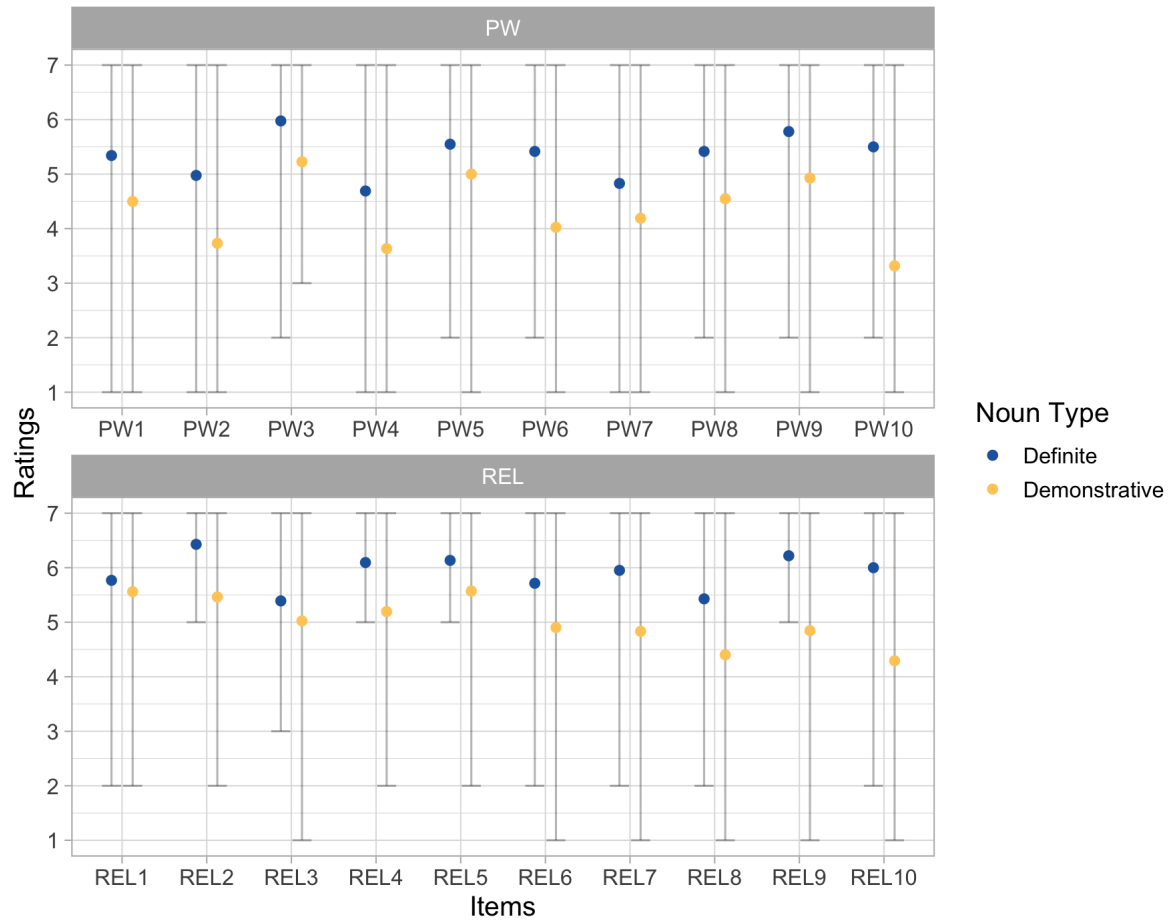


Figure 7: Study 2: Ratings as function of items, color-coded by bridged noun type (English)

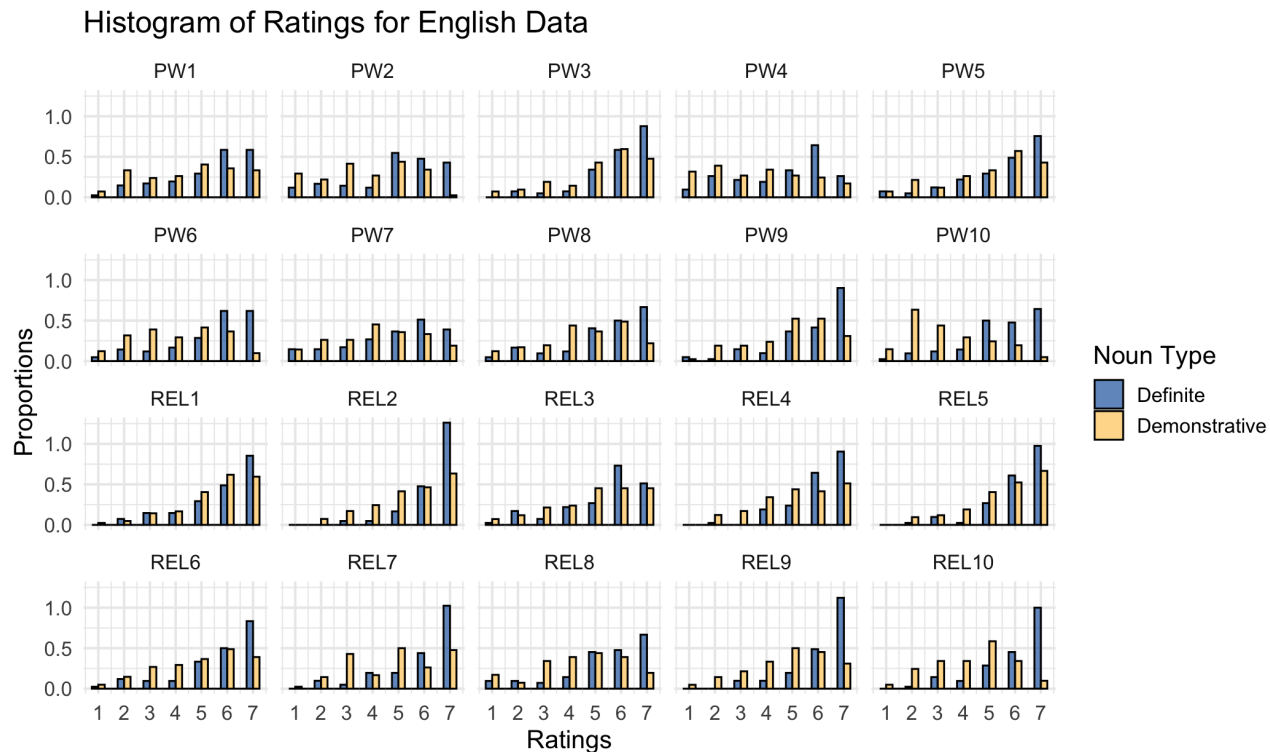


Figure 8: Study 2: Histograms of ratings by items, color-coded by bridged noun type (English)

Although these items did not reach statistical significance, Fig. 7 shows that, for these two items (REL1 and REL3), the average ratings for definite descriptions were still numerically higher than those for demonstrative descriptions, consistent with the other target items. Similarly, Fig. 8 shows that higher ratings were more frequently assigned to definite descriptions than to demonstratives. It is possible that a larger data sample would have revealed a significant effect for these items.

Finally, the comparison between target and control stimuli in English indicates that all bridging sentences received significantly higher ratings than the semantically or pragmatically odd control sentences. This may be due to several possible reasons. First, the difference between definite and demonstrative descriptions in English bridging might be less categorical for speakers. Second, participants may have taken the demonstrative description to have an affective reading (Lakoff 1974), accommodating definite-like uses. Lastly, participants may have tried to make their responses more fine-grained by varying their ratings according to the specific type of linguistic violations in the stimuli. To tease apart these possibilities, a follow-up study with a different empirical methodology would be necessary. We leave this for future research.

3.3.4 Cross-linguistic comparison

To compare the ratings across English and Mandarin, we included LANGUAGE as a fixed effect and performed two additional statistical tests.⁷ We fit a Cumulative Link Mixed Model to examine whether language interacts with bridged noun type within and across the bridging groups. Fur-

⁷We thank one of the anonymous reviewers for pointing us to add language as a factor and fit a bigger model.

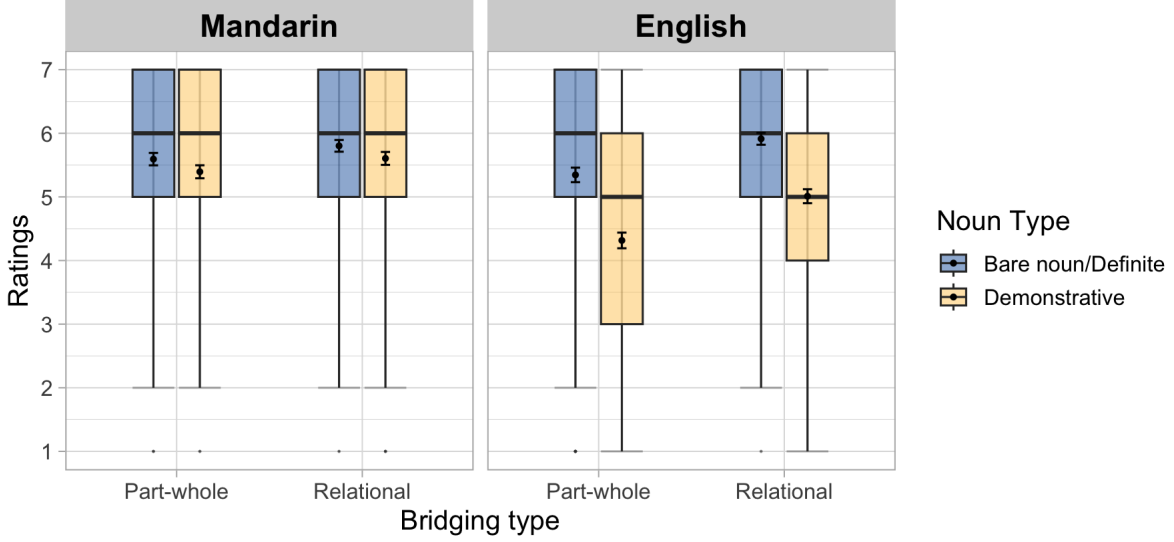


Figure 9: Ratings as function of bridging types, color-coded by bridged noun type

thermore, we conducted pairwise comparisons to compare the ratings by noun type across the two languages.

3.3.5 Statistical analyses

We fit a Cumulative Link Mixed Model using the ordinal package in R. The model included three fixed effects (LANGUAGE, BRIDGING TYPE, and BRIDGED NOUN TYPE) and two random effects (PARTICIPANT and ITEM) with both random intercepts and random slopes for NOUN TYPE. Model selection was based on AIC.

The modeling results showed a significant main effect of bridging type ($\beta = 0.88$, $SE = 0.16$, $z = 5.34$, $p = 9.48e - 08$) and noun type ($\beta = -1.43$, $SE = 0.16$, $z = -8.84$, $p < 2e - 16$), and a marginal main effect of language ($\beta = 0.26$, $SE = 0.16$, $z = 1.68$, $p = 0.09$). We found a significant interaction between language and bridging type ($\beta = -0.54$, $SE = 0.13$, $z = -4.05$, $p = 5.17e - 05$), and crucially, a significant interaction between **language** and **noun type** ($\beta = 1.13$, $SE = 0.16$, $z = 7.07$, $p = 1.59e - 12$). Fig. 9 plots these results.

To further confirm these interactions, we performed pairwise comparisons of the estimated marginal means (emmeans) using the emmeans package in R, applying the Tukey adjustment for multiple comparisons, for different combinations of **language** and **noun type** in part-whole and relational bridging, respectively.

In the part-whole bridging group, the pairwise comparison between Mandarin bare nouns and demonstratives showed no significant difference ($\beta = 0.30$, $SE = 0.16$, $z = 1.87$, $p = 0.24$), while the comparison between English definite and demonstrative descriptions revealed a significant difference ($\beta = 1.43$, $SE = 0.16$, $z = 8.84$, $p < 0.0001$). When comparing across languages, we found no significant difference between English and Mandarin definite descriptions ($\beta = -0.26$, $SE = 0.16$, $z = -1.68$, $p = 0.33$), but there was a significant difference between demonstratives in the two languages ($\beta = -1.39$, $SE = 0.16$, $z = -8.69$, $p < 0.0001$).

These patterns also held for relational bridging. The comparison between Mandarin bare nouns

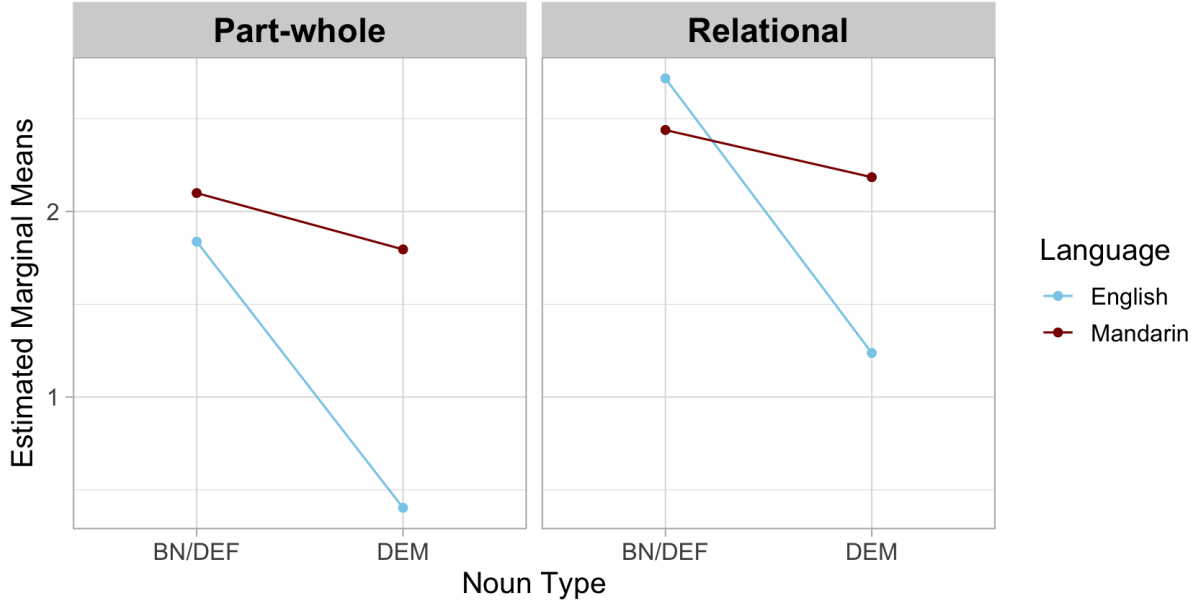


Figure 10: Estimated Marginal Means as function of noun types, color-coded by bridging type

and demonstratives showed no significant difference ($\beta = 0.25$, $SE = 0.16$, $z = 1.55$, $p = 0.41$). In contrast, there was a significant difference between English definite and demonstrative descriptions ($\beta = 1.48$, $SE = 0.16$, $z = 9.06$, $p < 0.0001$). Cross-linguistic comparisons revealed no significant difference between English and Mandarin definite descriptions ($\beta = 0.28$, $SE = 0.16$, $z = 1.76$, $p = 0.29$), but a significant difference between demonstratives ($\beta = -0.95$, $SE = 0.16$, $z = -5.90$, $p < 0.0001$).

Fig. 10 plots the Estimated Marginal Means as a function of noun types for each language, grouped by bridging type. For part-whole bridging, the lines remain separate. For relational bridging, the lines cross, due to the higher ratings for relational items than part-whole items in English.

3.3.6 Discussion

The statistical analyses indicate that, while English speakers show a strong preference for using definite noun forms over demonstratives in bridging sentences, this preference is less pronounced for Mandarin speakers. First, the significant interaction between language and noun type in the combined CLMM model suggests that the difference in ratings driven by noun type varies across languages. Specifically, the rating differences for Mandarin are significantly smaller than those for English. Second, the pairwise comparison shows that for each bridging group, demonstratives are rated significantly differently between the two languages, while definite noun forms are not.

These results underscore the distinct effect sizes observed in Studies 1 and 2. We find a clear difference between English and Mandarin: while demonstratives receive significantly lower ratings than definite descriptions in English bridging sentences, the significance of this rating difference in Mandarin is notably smaller compared to English.

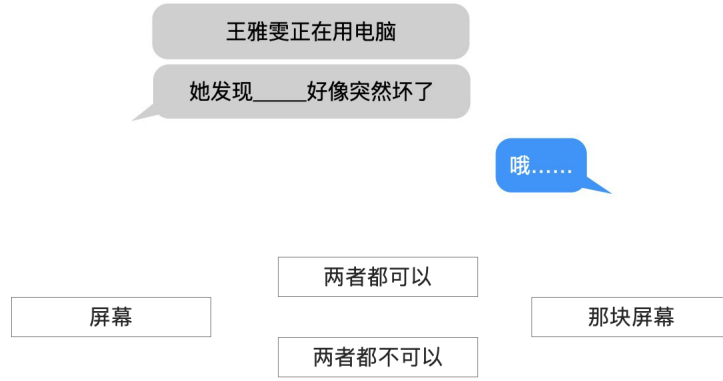


Figure 11: Sample item in Study 3 (Mandarin)

3.4 Study 3: Mandarin Production study

In order to further confirm the possibility of using the demonstrative *na* in bridging contexts, we conducted a production study where participants were shown a bridged context and asked to make a choice between bare noun and demonstrative. This study further confirms our finding from Study 1 that demonstratives allow both part-whole and relational bridging in Mandarin.

In this production study, we manipulated the linguistic context in a short message exchange to explore which nominal form(s), between bare nouns and demonstratives, are chosen by native Mandarin speakers. We closely followed the design of a comprehension task in Kroll 2020, Pratley, Pizarro-Guevara, Bleotu, Johnson, and Dillon 2024, and Pratley et al. in progress in designing this production study.

3.4.1 Methodology

Stimuli We designed 55 message exchanges with the following three components: (i) a background sentence that provided the linguistic context; (ii) a test sentence with a blank to be filled by a nominal; and (iii) a semantically uninformative reply (*oh*) that simply suggests the naturalness of the previous messages. Fig. 11 presents a sample item, which is glossed in (38).

- (38) a. *Wang Yawen zhengzai yong diannao*
 Wang Yawen PROG use computer
 ‘Wang Yawen is using the computer.’
 b. *ta faxian _____ haoxiang turan huai le*
 she find _____ seem.to suddenly break ASP
 ‘She found that _____ seemed to have broken all of a sudden.’
 c. *o ...*
 oh
 ‘Oh...’

Below each image stimulus, there were four text options for participants to choose from: a bare noun (e.g. *pingmu* ‘screen’), a demonstrative description (e.g. *na kuai pingmu* ‘that screen’), and options that read *both forms are acceptable* and *both forms are unacceptable*.

We included 24 target items, where the antecedent noun in the bridging relation was present in the background and the bridged noun was given in the options. The bridging type (part-whole vs. relational) was varied by manipulating the noun in the options. For example, in (38a), the background sentence introduces an antecedent noun *diannao* ‘computer’. Half the participants saw the options in (39a) (and (39c)), where the antecedent is in a part-whole relation with the bridged noun *pingmu* ‘screen’. The other half saw the options in (39b) (and (39c)), where the antecedent is the relatum argument with the relational bridged noun *chongdianqi* ‘charger’.

- (39) a. *pingmu* ‘screen’
 na kuai pingmu ‘that screen’
 b. *chongdianqi* ‘charger’
 na kuai chongdianqi ‘that charger’
 c. *liangzhe dou keyi* ‘both forms are acceptable’
 liangzhe dou bu keyi ‘both forms are unacceptable’

Control items In addition to the target items, we included 31 control items that did not involve bridging, divided into four groups. The first control group had a **Singular indefinite Noun** phrase in the background sentence (5 SN controls in total). The same noun was used in the options, when participants were asked to fill in the blank in the test sentence by selecting the acceptable nominal form(s).

- (40) *Wang Yawen zhengzai yong yi tai diannao; ta faxian {diannao, na tai diannao}*
 Wang Yawen PROG use one CL computer she find computer that CL computer
haoxiang turan huai le
 seem.to suddenly break ASP
 ‘Wang Yawen is using a computer. She found that _____ seemed to break all of a sudden.’

The second group of controls had a **Plural indefinite Noun** phrase instead in the background sentence (5 PN controls in total). Again, participants were asked to choose between the bare noun and demonstrative forms of the noun from the background sentence to fill in the blank in the test sentence.

- (41) *Wang Yawen zhengzai yong liang tai diannao; ta faxian {diannao, na tai diannao}*
 Wang Yawen PROG use two CL computer she find computer that CL
diannao } *haoxiang turan huai le*
 computer seem.to suddenly break ASP
 ‘Wang Yawen is using two computers. She found that _____ seemed to break all of a sudden.’

The third control group featured a bare noun in the background sentence. The bare **Noun** was **Repeated** in the test sentence, which characterizes the anaphoric use of a nominal. There were 12 RN controls in total.

- (42) *Wang Yawen zhengzai yong diannao; ta faxian {diannao, na tai diannao}*
 Wang Yawen PROG use computer she find computer that CL computer
haoxiang turan huai le
 seem.to suddenly break ASP
 ‘Wang Yawen is using the computer. She found that _____ seemed to break all of a sudden.’

The fourth and last group of controls had **No Nominal** antecedent in the background sentence (10

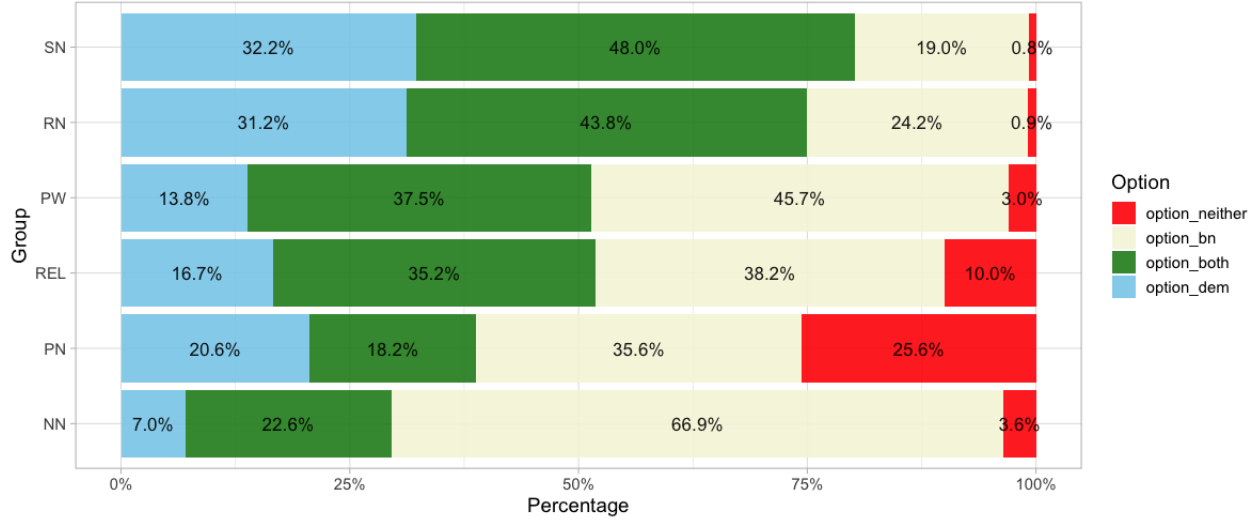


Figure 12: Rate of expression choice by condition in Study 3 (Mandarin)

NN controls in total). Participants were asked to fill in the blank in the test sentence with a novel noun, which was either bare or accompanied by a demonstrative. Note that NN controls should be distinguished from part-whole bridging with no linguistic antecedent in the background sentence, since in (43), one cannot uniquely identify a computer from the working situation.

- (43) *Wang Yawen ganggang zai gongzuo; ta faxian {diannao, na tai diannao}*
 Wang Yawen just.now PROG work she find computer that CL computer
haoxiang turan huai le
 seem.to suddenly break ASP
 ‘Wang Yawen was just working. She found that _____ seemed to break all of a sudden.’

We also included two attention checkers that asked participants to choose either ‘both forms are acceptable’ or ‘both forms are unacceptable’. This was to ensure that participants successfully loaded the images and paid attention to our instructions. A full list of stimuli used in Study 2 is included in Supplementary Materials.

Participants and procedure We recruited 100 native Mandarin speakers (age-balanced and gender-balanced) via Prolific. Participants were redirected to a Prolific survey, where they were asked to provide demographic and language background information. Participants were paid \$2–3 for their time.

Each participant was presented with 12 target items and 31 controls in randomized order. We adopted a within-subject design, but a participant saw only one variation (either part-whole or relational bridging) within each set of target items. Upon selecting one of the four options below the image stimuli, participants were directed to a new page to click *continue* to proceed to the next item.

3.4.2 Results

Fig. 12 plots the percentages of selecting each option in each condition. Below we report the results for the target items (PW (part-whole) and REL (relational) in the middle rows of the figure) and then the control items, including the baseline conditions (NN (no noun) and PN (plural noun) in the last two rows of the figure) and the anaphoric conditions (SN (singular noun) and RN (repeated noun) in the first two rows of the figure).

Target items For the target items, we calculated the percentages of participants who chose demonstratives as the bridged noun (i.e. who selected either `OPTION_DEM` or `OPTION_BOTH`): 51.3% chose demonstratives in part-whole bridging, and 51.9% chose demonstratives in relational bridging. For bare nouns, the percentages are 83.2% for part-whole bridging and 73.4% for relational bridging.

We conducted a McNemar’s test⁸ to examine the association between two factors: the bridging type (`PART-WHOLE` and `RELATIONAL`) and participants’ choice of a specific nominal form. For the latter factor, we collapsed the four options presented to the participants into two groups: the options indicating that participants chose demonstratives, `DEM_GOOD` (including `OPTION_DEM` and `OPTION_BOTH`), and the options indicating the reverse, `DEM_BAD` (including `OPTION_BN` and `OPTION_NEITHER`). The null and alternative hypotheses for the test are presented in (44).

- (44) a. **Null Hypothesis (H0):** The bridging type is not associated with participants’ choice. Participants’ choice of demonstratives does not vary as the bridging type varies.
 b. **Alternative Hypothesis (H1):** The bridging type is associated with participants’ choice. Participants choice of demonstratives varies as the bridging type varies.

The McNemar’s test showed a chi-squared statistic of 0.59867, with 1 degree of freedom. The p -value was 0.4391, greater than 0.05. Therefore, we failed to reject the null hypothesis. The statistical results therefore suggested that participants’ choice of demonstratives did not vary, in part-whole and relational bridging alike.

We further conducted a second McNemar’s test to examine the association between the bridging type and participants’ choice of bare nouns. The four options were collapsed differently, but again into two groups, based on whether participants chose bare nouns as the bridged noun form: `BN_GOOD` (including `OPTION_BN` and `OPTION_BOTH`) and `BN_BAD` (including `OPTION_DEM` and `OPTION_NEITHER`). In contrast to the test results for demonstratives, the McNemar’s test for bare nouns showed a chi-squared statistic of 212.42, with 1 degree of freedom. The p -value was $2.2e-16$, far less than 0.05. Therefore, we successfully rejected the null hypothesis. The results suggested a significant association between the bridging type and participants’ choice of bare nouns.

Taken together the percentages and the results from the McNemar’s tests, we concluded that (i) participants chose demonstratives as the bridged noun form in both types of bridging alike, but that (ii) participants chose bare nouns as the bridged noun form significantly more in part-whole bridging than in relational bridging.

Control items Table 1 presents the percentages of participants choosing demonstratives and bare nouns in the control conditions, respectively. Fewer participants chose demonstratives in the baseline conditions NN and PN, as compared to conditions with a linguistic antecedent that can be understood as a singular referent (SN and RN).

⁸We thank an anonymous reviewer for pointing out that the regular chi-square test (used in a previous version of the analysis) assumes that all observations are independent, but the within-subject design of our experiment results in non-independent observations because every participant was exposed to different conditions. Therefore, we switched to the McNemar’s test, a chi-square test that is applicable to non-independent observations.

Condition	Linguistic antecedents	% choosing DEM	% choosing BN
NN	none	29.6%	89.5%
PN	plural, ‘two CL NOUN’	38.8%	53.8%
SN	singular, ‘one CL NOUN’	80.2%	67%
RN	bare noun, ‘NOUN’	75%	68%

Table 1: Percentages of participants choosing DEM/BN in control items in Study 3 (Mandarin)

3.4.3 Discussion

The McNemar’s test results from the target items show that Mandarin speakers choose demonstratives in both part-whole and relational bridging to a similar extent. In contrast, bare nouns are more readily chosen in part-whole bridging than in relational bridging.

Our baseline conditions (NN and PN) confirmed that demonstratives are degraded without linguistic antecedents or with plural antecedents (although to different extent). In the NN condition, where no overt linguistic antecedent was provided, we predicted that bare nouns would be chosen the most. The prediction is borne out: most participants chose the ‘OPTION_BN’ response. In contrast, demonstratives across languages are known to prefer contexts where the antecedent is linguistically introduced (Patel-Grosz and Grosz 2017; Roberts 2002; Schwarz 2009). Indeed, the NN condition shows the lowest rate of ‘OPTION_DEM + OPTION_BOTH’ responses.

In the PN condition with plural bare nouns as antecedents, the highest rate of the ‘OPTION_NEITHER’ response is expected. However, the proportions of ‘OPTION_DEM’ and ‘OPTION_BN’ were higher than expected. The high choice of demonstratives might be due to other readings of *na*, which also allow deictic and affective readings (Himmelman 1996; Lakoff 1974; Potts and Schwarz 2010; a.o.). The high rate of bare nouns can be attributed to their ability to refer anaphorically, which we account for in the analysis in Section 4.

The difference in percentages of BN-choice between the PN-condition (53.8%) and the SN/RN-conditions (67% and 68%) might result from a singular bias of the nouns we used as the experimental items: even though all bare nouns are syntactically number-neutral and allow for both singular and plural interpretations, some bare nouns show a singular bias, preferring to be interpreted as semantically singular, while others show a plural bias, preferring to be interpreted as semantically plural. This property of bare nouns is experimentally attested in Modarresi (2014) and Law and Syrett (2017), both using pronominal anaphora as a proxy to identifying the contextual number bias of bare nouns. Specifically, a bare noun with a singular/plural bias prefers to be picked up by a pronoun with a matching number. In our results, the lowered percentages in the PN-condition can then be explained by the conflict between the semantic plurality marked by the numerals and the contextual singular bias of the bare nouns.

Lastly, the results in the SN and RN conditions confirmed the availability of the anaphoric use of both demonstratives and bare nouns. The high rate of demonstratives can be attributed to their anaphoric use. The high rate of bare nouns in anaphoric uses coincides with elicitation and corpus findings in previous works (Bremmers et al. 2022; Dayal and Jiang 2021).

4 Proposal

Taken as a whole, our results from the three studies suggest that Mandarin bare nouns and demonstratives readily allow both part-whole and relational bridging. Study 1 shows that in an offline sentence ratings task, Mandarin speakers accept bare nouns and demonstratives in both part-whole and relational bridging, though there is a marginal preference for bare nouns in both types of bridging. Study 3 shows that in production tasks, while speakers prefer bare nouns in part-whole bridging, they use demonstratives in both part-whole and relational bridging similarly. Thus, the general pattern surfacing is a preference for bare nouns in part-whole bridging, but demonstratives being licensed in both kinds of bridging.

Comparing our results from Mandarin to English, we observe in Study 2 and the subsequent language comparison a clear difference between Mandarin *na* and English *that*. Demonstratives are significantly degraded compared to definite descriptions in English, while this difference is significantly smaller in Mandarin.

How do existing analyses account for the pattern we present here? First of all, we note that our findings are not predicted by the analysis proposed in Dayal and Jiang 2021 because a) *na* does not behave like English *that*, and b) *na* is licensed in contexts that do not meet the anti-uniqueness presupposition that Dayal and Jiang assume for *na*'s denotation. While their analysis accounts for the more flexible use of demonstratives with the assumption that anti-uniqueness can be satisfied in extended situations, it is unclear exactly how situation extension can be constrained. Even if we adopt their assumptions of situations being extended through adding new discourse referents or events (see Saha et al. 2023 for an implementation of this idea), our stimuli did not systematically have these features that would license extending of situations.

The empirical pattern is also not fully compatible with the predictions of Jenks 2018. The main issue is that bare nouns and demonstratives are expected to be in complementary distribution given the principle *Index!* defined in terms of *Maximize Presupposition!*. In Jenks 2018, Mandarin *na* is analyzed as carrying a uniqueness-operator *iota* with an index, where the index is a presupposed predicate of type $\langle e, t \rangle$, while bare nouns only carry the *iota* operator without the index. *Maximize Presupposition!* (Heim 1991) requires that *na* be used over bare nouns when its presupposition is met — namely when the context supports anaphora and the index can be bound — and so the prediction of Jenks 2018 is that *na* would be required whenever anaphora is possible. So this proposal does not account for why both bare nouns and demonstratives are licensed in the two kinds of bridging, especially in contexts where anaphora is possible such as relational bridging. Recall that Dayal and Jiang (2021) and Bremmers et al. (2022) also show that Mandarin bare nouns do not behave like uniqueness-denoting definites. Thus, our study together with Bremmers et al. 2022 and Dayal and Jiang 2021 pose an empirical challenge for the *Index!* principle. That said, we show below that once we remove *Index!* from the analysis, the denotations Jenks proposes can be minimally modified to account for our data.

Taking stock, based on our results, we conclude that demonstratives are not anti-unique (cf. Dayal and Jiang 2021) or exclusively required for anaphoric uses (cf. Jenks 2018). The main data that need to be accounted for are as follows: a) bare nouns allow both part-whole and relational bridging; b) *na* allows both part-whole and relational bridging; c) there is a preference for bare nouns over demonstratives in part-whole bridging; d) Mandarin *na* differs from English *that* in allowing bridging more readily. In the next section, we propose a modified account of Mandarin bare nouns and demonstrative descriptions that can derive these empirical facts.

4.1 A new analysis of Mandarin definite expressions

The denotations of the two iota operators underlying the definite bare noun and the demonstrative description in Jenks 2018 are repeated below:

$$(45) \quad \begin{aligned} \text{a. } \llbracket \iota \rrbracket &= \lambda s_r. \lambda P_{\langle e, st \rangle} : \exists! x [P(x)(s_r)]. \iota x [P(x)(s_r)] \\ \text{b. } \llbracket \iota^x \rrbracket &= \lambda s_r. \lambda P_{\langle e, st \rangle}. \lambda Q_{\langle e, t \rangle} : \exists! x [P(x)(s_r) \wedge Q(x)]. \iota x [P(x)(s_r)] \end{aligned}$$

Following Jenks 2018, we assume that Mandarin nouns are cumulative predicates that contain both individuals and pluralities closed under sum (Link 1983; Schwarzschild 1996), and that classifiers take these as arguments and return atomic predicates (Chierchia 1998b), represented as AT .⁹

$$(46) \quad \begin{aligned} \text{a. } \llbracket N \rrbracket &= \lambda x \lambda s. P(x)(s) \\ \text{b. } \llbracket CL \rrbracket &= \lambda P \lambda x \lambda s. P(x)(s) \wedge AT(x) \end{aligned} \quad [\text{adopted from Jenks 2018:513}]$$

Our first divergence from Jenks 2018 is the removal of the *Index!* principle. Others have already shown that bare nouns in Mandarin allow both unique and anaphoric uses readily (Ahn 2019; Bremmers et al. 2022; Dayal and Jiang 2021). Our data in Studies 1 and 3 also show that bare nouns allow both part-whole and relational bridging. Following Schwarz 2009, we assume that part-whole bridging is licensed through situational uniqueness while relational bridging is licensed through an anaphoric reference to the relatum argument. That bare nouns allow relational bridging poses a further challenge to *Index!* because the principle would predict the non-indexed iota operator to be unavailable due to *Maximize Presupposition!*. Instead, we argue that both mechanisms of definite reference — namely uniqueness and familiarity — are available and subject to pragmatic competition, allowing a more gradient interaction between the expressions.

The second change is the modification of the denotation of the anaphoric definite in (45b). As shown in (45b), the domain restriction predicate Q is a predicate that holds of x , typeshifted from an individual-denoting item of type e . This can derive directly anaphoric uses of *na*, but not the bridging use because Q is a one-place predicate that holds of x , the bridged noun, directly. We propose to replace the domain restriction predicate Q with a possessive relation, whose content is contextually determined. In analyzing relational nouns and nouns that have turned relational, various possessive relation functions have been proposed in the literature (Barker 2011; Vikner and Jensen 2002; a.o.). For example, Vikner and Jensen (2002) propose a number of lexically-determined relation functions as well as a pragmatically-determined relation function *Prag*, as in (47), which can turn a non-lexically relational noun W into a relational noun that takes an individual y in addition to x and requires that y and x stand in some relation. Similarly, Barker (2011) discusses a type-shifting operator π defined in (48) that can take a non-relational noun and turn it into a relational one, where ‘ R is a free (pragmatically controlled) variable standing for the possession relation’ (p. 1114).

$$(47) \quad \text{Prag}(W) = \lambda y \lambda x [W(x) \ \& \ \text{related-to}(y)(x)] \quad [\text{Vikner and Jensen 2002:211}]$$

$$(48) \quad \pi = \lambda P \lambda x \lambda y. P(y) \wedge R(x, y) \quad [\text{Barker 2011:1114}]$$

We propose to build the denotation of this relationalizer (specifically π) to the meaning of *na* to make it a relationalizing operator as in (49).¹⁰

⁹We have simplified the denotations and removed the kind vs. individual distinction (Dayal 2004), and refer the reader to Jenks 2018 for more discussion.

¹⁰We thank our anonymous reviewer for suggesting this connection.

$$\begin{aligned}
(49) \quad \llbracket na \rrbracket &= \lambda s_r. \lambda P_{\langle e, st \rangle}. \lambda z : \iota x [\pi(P)(z)(x)(s_r)] \\
&= \lambda s_r. \lambda P_{\langle e, st \rangle}. \lambda z : \iota x [P(x)(s_r) \wedge R(z, x)(s_r)]
\end{aligned}$$

In this analysis, *na* takes a resource situation s_r , a predicate P and a variable z as arguments and returns the unique entity x that is P in s_r and stands in some relation R with z in s_r . For R , we follow Vikner and Jensen 2002 and Barker 2011 in assuming that it is either a generic ‘related-to’ predicate or a free variable for some possessive relation.

We leave the denotation for non-indexed ι in (45a) as is and assume that bare nouns receive a definite interpretation through uniqueness without a slot for an index. Our analysis for *na* also does not diverge from that of Jenks 2018 extensively. A reviewer suggests that the analysis can be made even more similar to that in Jenks 2018 if we assume that the original domain restriction predicate $Q(x)$ in (45b) is thought of as optionally allowing a relational shift of the form π in addition to IDENT. Then, the only change necessary is the assumption that when the indexed ι takes an index as an argument, it can be turned into a property through IDENT ($\lambda x. x = g(i)$) or through π ($\lambda x. \pi(x, g(i))$). The only difference between this suggested analysis and our proposal in (49) is where the relational predicate is given: in our analysis, *na* directly takes an index as its argument and provides a relational predicate in its denotation; in the modified Jenks 2018 analysis, the index first turns into a predicate through a relational shifter and combines with *na*. While the two do not seem to have different empirical consequences, we choose to proceed with the analysis we propose in (49) because a) it parallels the semantics of anaphoric definites proposed in Schwarz 2009 in that it directly takes an index rather than a contextual predicate and b) it keeps the relational predicate in the asserted content rather than in the presupposed domain. The addition of Q in the presupposition was motivated in Jenks 2018 to explain *Index!* through *Maximize Presupposition!*, but this is not necessary for our analysis. Furthermore, we have reasons to think that the index argument of an anaphoric definite should be at-issue and not backgrounded. It has been argued that the index information of anaphoric, referential definites should be at-issue (Gutzmann and McCready 2014) or at least can be at-issue (Ebert et al. 2020). For example, in the famous misrepresentation cases such as *The man drinking a martini is tall* where the intended referent is not drinking a martini (Donnellan 1966), Gutzmann and McCready (2014) argue that what is at-issue is the index information and not the description. That the bridged content is at-issue and can be directly negated is also shown in (50).

- (50) A: There was a book reading event here yesterday.
 B: Yes I know. In fact, I was here and met the author.
 A: No you didn’t. The person you met was not the author of that particular book but another book that was on sale yesterday.

In B’s utterance, *the author* is intended to refer to the author of the book featured in the book reading event. However, it turns out that who B met was an author of a different book. If $Q(x)$ is backgrounded, B’s statement in (50) should not be directly rejectable given that the intended referent is in fact an author. However, A can directly reject B’s claim based on the fact that the book-author relation does not hold between the author B met and the book featured in the event. Given these, we proceed with (49), which bakes the relational information into the denotation of *na* and places it in the asserted component. Below, we show how our analysis can derive the data presented in our empirical studies.

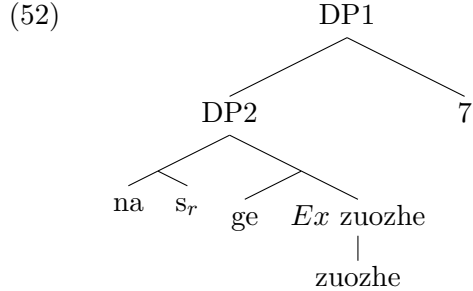
4.2 Deriving the data

na in relational bridging

We start with *na* in relational bridging. In Studies 1 and 3, we showed that *na* is licensed in contexts of relational bridging. Note that we specifically made use of relational nouns for the stimuli. Since *na* in (49) is defined to take a non-relational noun and turn it into a relational noun, we need to further assume a detransitivizing type-shifter *Ex*, as defined in (51), which applies to a relational noun before the noun composes with the rest of the DP structure.¹¹

$$(51) \quad Ex = \lambda R \lambda x \exists y R(x, y) \quad [\text{Barker 2011:1114}]$$

This kind of type-shifter is independently motivated for licensing relational nouns without overt possessors and for interpreting a relational noun with an overt possessor in a pragmatically-defined way rather than the lexically-defined way. For example, Barker (2011) discusses that the phrase *John's brother* does not only have the lexically-defined reading where it refers to a person who is born as John's brother, but also has a pragmatically-licensed meaning, referring to 'some male person who has a sibling and who is related to John through some kind of circumstantial association' (p. 1115).¹² For the latter reading, Barker gives the denotation $\pi(Ex(\llbracket \text{brother} \rrbracket)) = \lambda x \lambda y. \exists z \text{brother}(z, y) \wedge R(x, y)$. Given how readily π and *Ex* can apply to shift the denotation of nouns between relational and non-relational, we assume that *Ex* is freely available for a relational noun when it combines with *na*. The *R* can be contextually provided, or can pick up the lexically-determined relation from the noun, thus allowing both pragmatic and lexical relation to hold between *x* and *z*. For the *book-author* example, *na ge zuozhe* ('that author') would receive the following denotation, which would be translated to English as 'the unique individual *x* such that *x* is an author of some *y* and stands in some relation to *g(7)*'.



$$(53) \quad \begin{aligned} \llbracket \text{zuozhe} \rrbracket^g &= \lambda z \lambda x \lambda s. \text{author}(z, x)(s) \\ \llbracket Ex \text{ zuozhe} \rrbracket^g &= \lambda x \lambda s. \exists y. \text{author}(y, x)(s) \\ \llbracket ge \text{ } Ex \text{ zuozhe} \rrbracket^g &= \lambda x \lambda s. \exists y. \text{author}(y, x)(s) \wedge AT(x) \\ \llbracket DP2 \rrbracket^g &= \lambda z. \iota x [\exists y. \text{author}(y, x)(s_r) \wedge AT(x) \wedge R(z, x)(s_r)] \\ \llbracket DP1 \rrbracket^g &= \iota x [\exists y. \text{author}(y, x)(s_r) \wedge AT(x) \wedge R(g(7), x)(s_r)] \end{aligned}$$

¹¹ As an anonymous reviewer suggests, there is an alternative analysis for this, which is to simply assume that the possessor is base-generated in the argument position of the noun and moved to the higher position, where it can also serve as an argument of the poss operator. However, as Schwarz (2009) notes, there are distinctions observed between relational nouns that carry overt possessor arguments inside the noun and those that do not and are assumed to carry indices higher up. Because it is unclear how to associate the index taken by *na* with the internal noun index, we assume the *Ex* detransitivizer for our paper. We further note that the use of *Ex* is necessary regardless of our choice here given that non-relational nouns can take on relational meanings and relational nouns can take meanings that are not lexically encoded in their semantics.

¹² In case this use is hard to imagine for some readers: suppose a family has three brothers — Andy, Billy, and Cindy — and John and two of his friends must each pair up with one of the brothers to dance. Then the description *John's brother* may refer to the brother who pairs up with John.

If interpreted lexically, then R would pick up the author-book relation and return the unique x such that x is an author of something and is an author of the book $g(7)$. If interpreted pragmatically, then R can pick up some salient relation and return the unique x such that x is an author of something and stands in some relation to $g(7)$. This can be applied to all of our relational stimuli.

One prediction of this analysis is that *na* should be able to take a non-relational noun as its argument and turn it into a pragmatically-relational noun that stands in some relation to the index argument z . We test this prediction in Study 4 presented in Section 5 and show that this prediction is borne out.

***na* in part-whole bridging**

The analysis for *na* above extends to part-whole bridging straightforwardly, with R being specified as the ‘part-of’ relation. For the pair *bike-seat*, for example, *na ge chezuo* (‘that seat’) would return the unique x that is a seat in s_r and stands in some relation to z . In the given context where the first sentence mentions the speaker’s bike, z would map to that bike, and R would most naturally map to the ‘part-of’ relation.

- (54) $\llbracket \text{na ge chezuo } 7 \rrbracket^g = \iota x [\text{seat}(x)(s_r) \wedge AT(x) \wedge R(g(7), x)(s_r)]$
- a. $g(7)$: the bike
 - b. R : related to (‘part-of’)

However, given that the bare noun can resolve the part-whole bridging with simpler semantics that only involves situational uniqueness, we would expect the use of *na* in part-whole bridging to be redundant and thus less preferred.

Mandarin vs. English

Before moving onto bare nouns, we briefly discuss why English demonstratives would not allow such bridging uses under our analysis. For demonstratives to allow bridging cases as discussed here, they must carry an index but have a relational semantics, as in (49). Recall, however, that this relational semantics is an extension of strong definites in Schwarz 2009. We follow Schwarz 2009 and assume that the English definite determiner has both uniqueness-based and familiarity-based interpretations, and that the corresponding bridging cases are available by extension. Demonstratives are often analyzed as carrying an additional restriction in comparison to the definite (Elbourne 2008; King 2001; Robinson 2005), and Ahn (2022) and Blumberg (2020) argue that demonstratives are blocked by economy principles when definites are available. Thus, we expect demonstratives to not allow bridging uses because definite determiners can do the job. Additional support for this pragmatically-derived competition comes from the fact that demonstratives in English do allow deferred reference that seems to rely on relational semantics as in (55) (see Nunberg 1993 for more discussion on deferred reference).

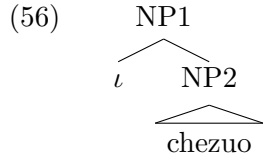
- (55) I liked that movie (pointing to a poster).

A possible analysis of (55) is that the demonstrative takes the poster as its demonstratum but refers to the unique entity (the movie) that is related to the poster by some function. Deferred use of demonstratives in English is restricted to deictic cases, where the speaker points to the entity. Since definite descriptions do not allow deixis to enter the restrictive content (Ahn and Davidson 2018; Ebert et al. 2020), we correctly predict that demonstratives would allow such relational usage. In non-deictic cases, however, the definite description is always available, thus making demonstratives redundant. Analyzing the bridging use of demonstratives to be available in principle but blocked due

to competition with definite articles is compatible with our data in Study 2 where demonstratives are significantly degraded in bridging uses but still rated higher than pragmatically or semantically odd sentences.

Bare nouns in part-whole bridging

For bare nouns, part-whole bridging is straightforwardly accounted for using the same logic in Jenks 2018. We assume that ι typeshifts the predicative noun into an individual-denoting entity by feeding it a situation variable and returning the unique entity that meets the noun denotation. The situation can be bound by the topic situation, which in turn can be fixed to the relevant question under discussion (Schwarz 2009), licensing part-whole bridging.

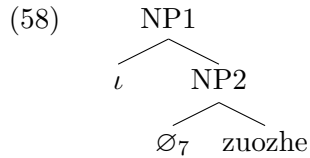


(57) $\llbracket \text{NP1} \rrbracket^g = \llbracket \iota \text{ chezuo} \rrbracket^g = \iota x.seat(x)(s_r)$

Bare nouns in relational bridging

Finally, the licensing of bare nouns in relational bridging is not straightforwardly predicted by Jenks 2018, but we argue that the denotation in (45a) can indeed derive relational bridging cases as well if we assume that the possessor argument is covertly available in the head noun in Mandarin. The mechanism licensing relational bridging with strong definites in Schwarz 2009 is the possessor index that is introduced outside the complement noun. There is, however, another way to introduce this possessor argument, which is through the head noun that contains the possessor argument. For example, Schwarz (2009) finds that while the anaphoric definite article appears for relational bridging in German, if the relatum argument is overtly expressed inside the complement of the head noun, the unique definite article is licensed instead.

Adopting this intuition, we argue that the bare nouns that license relational bridging contain the relatum argument as a covert variable. This variable provides an argument to the relational noun internally before it combines with the ι operator, bleeding the need for a relational demonstrative that introduces an external index. This is possible in Mandarin because Mandarin more readily allows argument drop than languages like English and German (Huang 1984; Liu 2014; a.o.).



(59) $\llbracket \text{NP2} \rrbracket^g = [\lambda z \lambda x \lambda s.author(z, x)(s)](g(7))$
 $\llbracket \text{NP1} \rrbracket^g = \iota x.author(g(7), x)(s_r)$

4.3 Prediction: relationality

Note that all of our relational bridging stimuli in Studies 1 and 3 were lexically relational nouns, which are argued to take two argument slots. The analysis for bare nouns in relational bridging above is only possible if the noun itself carries a relatum argument inside its denotation and does not need an external index to be hosted by the demonstrative *na*. In other words, we expect relational

bridging to be only possible for bare nouns if the predicate is lexically relational. Together with the analysis for *na*, we predict that a non-lexically relational noun would license bridging only with *na* and not with the bare noun. Testing this prediction is crucial in strengthening our analysis because we argue that the uniqueness-based analysis of definite bare nouns in Jenks 2018 can be maintained to account for what looks like an anaphoric reference on the surface. Thus, we tested our prediction by designing another sentence ratings task (Study 4) that investigated the availability of bridging of (non-)relational nouns with bare nouns and demonstratives. Our analysis predicts that demonstratives would license bridging regardless of whether the noun is lexically relational or not, since the demonstrative *na* itself is a relationalizer that introduces an external relatum argument. On the other hand, bare nouns are predicted to only license bridging if the noun is lexically relational. Otherwise, the bare noun would not have a slot to host the relatum argument that licenses bridging. This prediction is borne out. In the next section, we present our data.

5 Study 4: Testing the predictions of the proposal

5.1 Methodology

This sentence ratings task explored whether bare nouns and demonstratives in Mandarin license relational bridging when the bridged noun is lexically relational or non-lexically relational. Recall that all of our relational bridging stimuli in Studies 1 and 3 were lexically relational nouns, thus not allowing us to test whether lexical relationality is what allowed relational bridging for bare nouns. We adopted a 2×2 design, manipulating two factors: NOMINAL TYPE (bare nouns vs. demonstratives) and RELATIONALITY (relational vs. non-relational) of the bridged noun.

Target stimuli We included 6 groups of target stimuli (24 sentences in total) evenly distributed in NOMINAL TYPE and RELATIONALITY. For example, one representative group of stimuli is presented in (61), where the indefinite antecedent *xiaoshuo* ‘novel’ is associated with the relational noun *zuozhe* ‘author’ or the non-relational noun *xiaoshuo-jia* ‘novelist’. We manipulated the nominal type of these bridged nouns via the presence and absence of the demonstrative *na* and a classifier before the noun.¹³

- (61) *Wang Yawen xihuan shoucang tushu. ta mei-ci zhao-dao le yi ben xihuan de*
 Wang Yawen like collect book. she every-time find-arrive ASP one CL like DE
xiaoshuo, *zuihou dou hui faxian ziji du-guo* {(na wei) *zuozhe*, (na wei)
 novel, finally always will discover self read-pass that CL author that CL
xiaoshuo-jia} *xie de ling yi ge gushi*.
 novel-person write DE another one CL story

¹³As noted by an anonymous reviewer, our stimuli for Study 4 involved covarying uses while our stimuli in previous studies involved intersentential cases. We assume that both intersentential and covarying uses would involve anaphora as argued in Schwarz 2009 and do not predict this to change the results. For example, the intersentential version of (61) is shown in (60), and here, we find the same pattern as we did for the test items: the non-relational bare noun *xiaoshuo-jia* (‘the novelist’) is much less natural than the demonstrative form *na wei xiaoshuo-jia* (‘that novelist’).

- (60) *Wang Yawen xihuan shoucang tushu. ta zhao-dao le yi ben xihuan de xiaoshuo. ta zuihou faxian*
 Wang Yawen like collect book she find-arrive ASP one CL like DE novel she finally discover
ziji du-guo {(na wei) *zuozhe*, (na wei) *xiaoshuo-jia*} *xie de ling yi ge gushi*.
 self read-pass that CL author that CL novel-person write DE another one CL story
 ‘Wang Yawen likes to collect books. She found a novel that she likes. She eventually realizes that she has
 read another story written by the author/the novelist.’

‘Wang Yawen likes to collect books. Every time she finds **a novel** that she likes, she eventually realizes that she has read another story written by **the author/the novelist**.’

In designing these noun pairs, we adopted two strategies presented in Schwarz 2009. The first strategy used for the first three examples was simply coming up with minimal pairs of synonyms where one was relational and the other was not, like ‘author’ and ‘novelist’. The second strategy was that of compounding. Schwarz (2009) notes that if the relatum argument is overtly present and forms a compound with the relational noun, then the noun is no longer relational. Using this strategy, we created noun pairs by taking a relational noun and making a compound with an overt relatum argument. An example of this is ‘author’ vs. ‘fairy.tale-author’. The complete list of the target stimuli can be found in Supplementary Materials.

To decide whether or not a noun is lexically relational, we extended the *of* diagnostic in Barker 2011. As discussed above, in English, relational nouns can appear with an *of*-possessive, while non-relational nouns cannot. This contrast is demonstrated in (62).

- (62) a. the author of a novel
b. *the novelist of a novel

We extended this diagnostic to Mandarin nouns. Specifically, Mandarin possessive constructions are expressed with a genitive marker *de*, as shown in (63), and only relational nouns (e.g. *zhuren* ‘owner’ in (63a)) are allowed after *de* in this construction, in contrast to non-relational nouns (e.g. *ren* ‘person’ in (63b)).

- (63) a. *yi zhi mao de zhuren*
one CL cat DE owner
‘the owner of a cat’
b. **yi zhi mao de ren*
one CL cat DE person
‘the person of a cat’

Based on this diagnostic, we conducted a small-scale norming study ($n = 8$), asking native speakers to rate the naturalness of Mandarin *de*-constructions with the nouns in our stimuli. For example, in (64), *zuozhe* ‘author’ received a high naturalness rating and was labelled as relational, while *xiaoshuo-jia* ‘novelist’ received a low rating and was labelled as non-relational.

- (64) a. *na ben xiaoshuo de zuozhe*
that CL novel DE author
‘the author of that novel’ [Average rating: 4.88/5]
b. **na ben xiaoshuo de xiaoshuo-jia*
that CL novel DE novelist
‘the novelist of that novel’ [Average rating: 1.50/5]

We present the list of nouns used in our stimuli in Table 2. In the norming study, the relational nouns received an average naturalness rating of 4.5 or above, while the non-relational nouns received an average naturalness rating of 1.5 or below.

Control stimuli In addition to the target stimuli, we included 12 pairs of control stimuli (24 sentences in total): 4 pairs of BOTHGOOD stimuli where the linguistic context does not prefer either BN or DEM, 4 pairs of BNGOOD stimuli where the linguistic context requires a generic reading or a unique reading of the nominal and therefore BN is preferred, and 4 pairs of DEMGOOD stimuli where

Antecedent	Relational noun	Non-relational noun
novel	author	novelist
painting	creator	artist
solo concert	performer	musician
song	singer	jazz singer
literary work	author	fairy tale author
restaurant	owner	diner owner

Table 2: Stimuli of Study 3

the linguistic context requires an anti-unique reading or includes a pre-nominal noun modifier and therefore DEM is preferred. Taken together, the control stimuli can be divided into 5 groups. The complete list of the control stimuli can be found in Supplementary Materials.

Participants and procedure We recruited 100 native Mandarin speakers (age-balanced and gender-balanced) via Prolific. Participants were redirected to a PCIBex survey, where they were asked to first provide some demographic and language background information and then complete the ratings task. Participants were paid \$2–3 for their time.

Each participant was presented with a total of 18 pairs of stimuli (36 sentences in total): 6 pairs of target stimuli and 12 pairs of control stimuli. The presentation order within each pair was pseudo-randomized, and the presentation order of all of the pairs was randomized. The experiment had a between-subject design: each participant saw only the relational condition or only the non-relational condition. For each trial, participants were shown two sentences in a pair side by side (following Marty et al. 2020; Bryant 2022) and asked to rate the naturalness of these sentences on a 7-point Likert scale, as demonstrated by the sample stimuli in Fig. 13.

5.2 Results

Target stimuli We fit a Cumulative Link Mixed Model via the ordinal package (Christensen 2022; Bross 2019) in R to compare ratings in different conditions. Our model included two fixed effects (NOMINAL TYPE and RELATIONALITY) and two random effects (PARTICIPANT and ITEM).

The modeling results showed a main effect of **nominal type** ($p < 0.001$) and **relationality** ($p < 0.05$), as well as a significant interaction between the two factors ($p < 0.001$). The interaction justified a subset analysis: we fit a Cumulative Link Mixed Model for each relationality condition. In the relational group, NOMINAL TYPE is not significant ($p = 0.732$), while in the nonrelational group, NOMINAL TYPE has a main effect ($p < 0.001$).

Fig. 14 plots the average ratings in each of the four conditions. On average, relational bare nouns were rated 5.37; relational demonstratives were rated 5.2; non-relational bare nouns were rated 4.82; non-relational demonstratives were rated 5.48. In addition, Fig. 15 plots the distribution of ratings in each condition.

Control stimuli We fit a Cumulative Link Mixed Model for each group of the control stimuli. Our model included one fixed effect (NOMINAL TYPE) and one random effect (PARTICIPANT). For two of the groups (BNGOOD_GENERIC and DEMGOOD_ANTIUNIQUE), the modeling results showed a singular fit, which suggested that the random effect was redundant, and thus motivated us to fit a Cumulative Link Model with no random effect considered. In all five groups of control stimuli, we



请给句子的自然程度打分，1分为最不自然，7分为最自然。

王雅雯喜欢收藏图书。她每次找到了一本喜欢的小说，最后都会发现自己读过那位小说家写的另一个故事。

最不自然 1 2 3 4 5 6 7 最自然

王雅雯喜欢收藏图书。她每次找到了一本喜欢的小说，最后都会发现自己读过小说家写的另一个故事。

最不自然 1 2 3 4 5 6 7 最自然

继续

Figure 13: Sample stimuli in Study 4 (Mandarin)

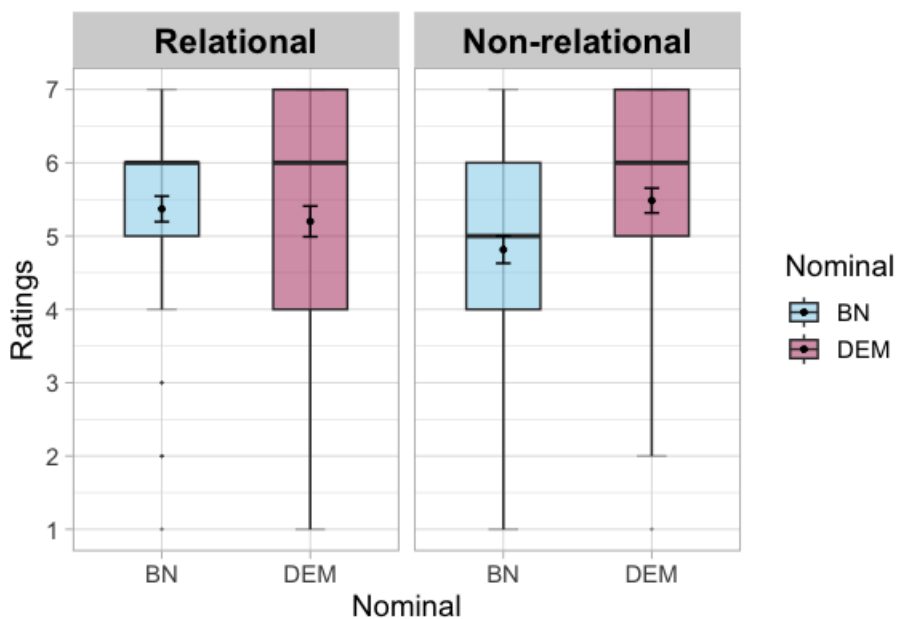


Figure 14: Ratings as function of nominal type (bottom & box color), grouped by relationality (top) in Study 4 (Mandarin)

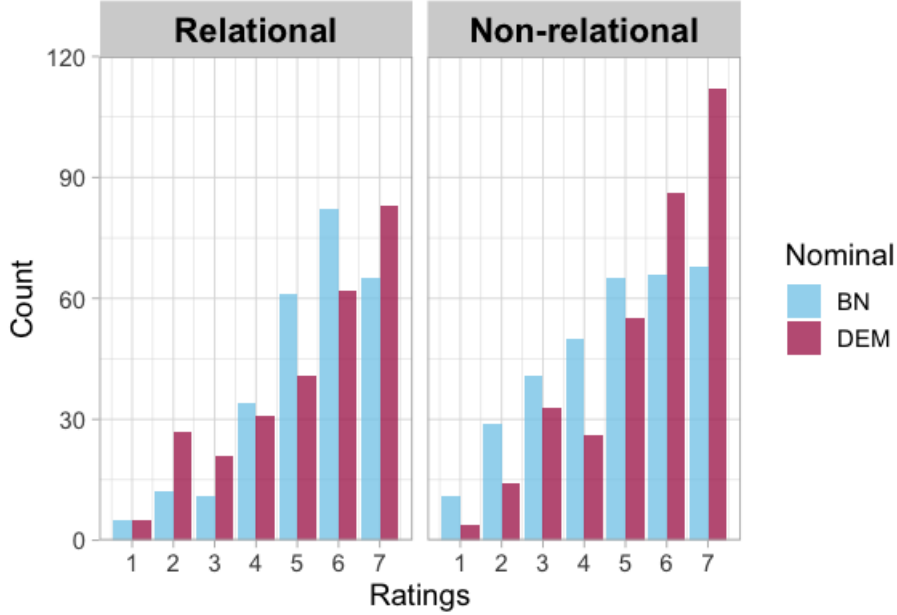


Figure 15: Distribution of ratings by nominal type (color) and relationality in Study 4 (Mandarin)

Group	BN	DEM
BOTHGOOD	6.07	4.93
BNGOOD_GENERIC	6.40	2.89
BNGOOD_UNIQUE	6.22	3.60
DEMGOOD_ANTIUNIQUE	3.15	6.14
DEMGOOD_NOUNMODIFIER	4.03	6.32

Table 3: Average ratings of control stimuli in Study 4 (Mandarin)

found a main effect of **nominal type** ($p < 0.001$). The average ratings for each group and condition are summarized in Table 3. The box plots that visualize the average ratings for each group can be found in the Appendix.

5.3 Discussion

Our results show that bare nouns and demonstratives differ in their ability to license bridging with non-relational nouns. For lexically relational nouns, both bare nouns and demonstratives are licensed for bridging. For non-relational nouns, however, demonstratives readily license bridging while bare nouns are significantly degraded. The results thus confirm the prediction of our analysis, where bare nouns rely on the relational predicate to be equipped with the relatum index internally while demonstratives can add their own external index to license relational bridging.

As for the control stimuli, our results show a general preference for bare nouns over demonstratives in the BOTHGOOD condition. This preference can be explained by a pragmatic competition between the two definite forms, where bare nouns are simpler in form and thus more economical than demonstratives (see the discussion in Section 6.1.1 on how to derive their competition). We

further observe that when nominals have a generic reading or refer to an inherently unique entity, bare nouns are preferred. In contrast, when the linguistic context establishes multiple possible referents or contains a pre-nominal noun modifier, demonstratives are much preferred. These results confirm the predictions of existing literature on bare nouns and demonstratives.

6 Conclusion

Studies 1 through 4 together draw the following empirical picture. For part-whole bridging which is argued to be based on situational uniqueness, Mandarin bare nouns and demonstrative descriptions are licensed, though there is a preference for bare nouns. For relational bridging involving lexically relational nouns, bare nouns and demonstratives are both licensed. For relational bridging involving non-lexically relational nouns, demonstratives are readily licensed while bare nouns are degraded. For English, demonstratives are significantly degraded in comparison to definite descriptions for both part-whole and relational bridging.

These observations are accounted for by our analysis of Mandarin bare nouns and demonstratives. For bare nouns, we adopt the analysis of Jenks 2018 that they are uniqueness-denoting expressions that cannot take an external index. Part-whole bridging is derived readily because all it requires is uniqueness in some relevant situation. Relational bridging with lexically relational nouns is also possible because the nouns themselves bring a relatum argument which can be covert in Mandarin. Relational bridging with non-relational nouns is generally unavailable for bare nouns because there is no slot for a relatum argument. For the demonstrative *na*, we argue that it is a relationalizer, introducing a relational predicate in the restriction of the ι . This analysis differs from that of Jenks 2018 in that a) the index argument is now a relatum argument introduced by the relational predicate, and b) the relational predicate is in the restriction of ι rather than in the presupposed content. Part-whole bridging is possible with demonstratives because the salient relation R can be specified to be part-whole, though the simpler semantics of bare nouns would be preferred in general. Relational bridging is possible for *na* regardless of whether the noun is lexically relational or not because the external index serves as the relatum argument that stands in some salient (lexical or pragmatic) relation with the referent of the demonstrative description. In the remainder of this paper, we discuss the main implications of our study.

6.1 Implications

6.1.1 Mandarin definite expressions

Definite bare nouns denote uniqueness. Our analysis has implications that contribute to the debate on how Mandarin divides up the definite space morphosyntactically. As discussed above, how Mandarin definite expressions map onto different meanings of definiteness has been debated in the literature. Our view aligns better with that of Jenks 2018 which argues that bare nouns track with unique definites and demonstrative descriptions track with anaphoric definites, though with some crucial differences. We agree with Jenks that definite bare nouns should be analyzed as requiring uniqueness and not carrying an external index. This restriction has been contested in the literature, with anaphoric uses of bare nouns being reported (Ahn 2019; Bremmers et al. 2022; Dayal and Jiang 2021). However, while this can be seen as an argument against the *Index!* principle which requires bare nouns to be blocked whenever the demonstrative is licensed, that bare nouns are licensed in anaphoric contexts is not an argument against the analysis of bare nouns as uniqueness-denoting definites for at least two reasons. First, it is easy to construe anaphoric contexts as involving smaller situations where uniqueness holds (Elbourne 2005), thus recasting ‘anaphoric’ uses of bare nouns as

‘unique’ uses in smaller domains. Second, our data where non-relational nouns cannot be bridged with bare nouns cannot be explained if bare nouns can also carry indices.

***na* is a relational determiner.** We diverge from Jenks 2018 for the analysis of *na*. Instead of arguing that it is directly anaphoric to some given entity, we broaden the notion to that of ‘relational anaphora’, where an index is introduced as Jenks argues, but is not predicated directly to the referent but instead added as an argument to a contextually-salient relation. Thus, *na* behaves as a nominal possessive in our analysis, and not as an anaphoric definite. This view of demonstratives aligns with existing analyses of demonstratives that have relational semantics (Elbourne 2008, Ahn in press).

Deriving the competition between bare nouns and demonstratives. How does this account derive the different distributional patterns of the bare noun and the demonstrative in Mandarin? Instead of *Index!*, we propose that the two expressions compete under a pragmatic principle of economy when they are truth-conditionally equivalent. Such pragmatic principles have been motivated independently in other works (Meyer 2014; Schlenker 2005) to account for competition between definite expressions (see Ahn 2019; Blumberg 2020). The idea is that if the bare noun and the demonstrative description refer to the same individual, then the bare noun is pragmatically ranked higher because it is semantically simpler. One important difference between our analysis and that of Jenks 2018 is that the relational predicate is inside the restriction of the ι operator in our account, rather than being a presupposition. This allows *na* and the bare noun to be subject to pragmatic competition like *Minimize Restrictors!* (Schlenker 2005), which derives different distributional patterns of the two expressions.

The pragmatic competition can explain why in examples like (65) the demonstrative description is degraded when referring to the unique moon.¹⁴

- (65) (#*na ge*) *yueling sheng shang lai le*.
 that CL moon has-risen
 ‘The moon has risen’ [Jenks 2018:507]

We do not argue that the degradedness of *na* here is due to some lexical property such as anti-uniqueness (Dayal and Jiang 2021). Instead, we assume that both *na* and the bare noun are in principle available to resolve the referent but compete for pragmatic economy. In (65), the bare noun resolves the referent through situational uniqueness, referring to the unique moon in a given situation. The demonstrative can resolve to the same referent if *R* can be set as a location relation with *z* being the relevant location under discussion. While both expressions resolve to the same entity, the demonstrative description has one additional restriction, namely the *R* relation. In discussing the distribution of definite and demonstrative descriptions in English and French, Robinson (2005) argues that the additional restriction that demonstratives carry lead to an anti-uniqueness (‘non-maximality’ in the paper) inference. Schlenker (2005) proposes that redundant restrictions of a definite description are blocked unless they have a pragmatic function. For example, ‘my tall brother’ is degraded if the speaker has only one brother and tallness does not serve a pragmatic purpose in the conversation. We follow the intuitions in Robinson 2005 and Schlenker 2005 and argue that the demonstrative is degraded in (65) because the *R* restriction is redundant.

¹⁴We thank an anonymous reviewer for bringing up this example.

6.1.2 Unifying the two variants of the strong article in Schwarz 2009

In Schwarz 2009, two versions of the strong definite article are proposed to account for the directly anaphoric use and the bridging use. That there is a relational variant of the strong article is left as a future problem to be investigated (Schwarz 2009:142). In our analysis, *na* is analyzed as a relational element, with direct anaphora being one of its possible uses where the salient relation R is that of identity. This aligns with the view in Clark 1975 where identity is assumed to be the shortest bridge that can be drawn between the antecedent and the referent of the bridged expression and how Elbourne (2008) implements the semantics of English demonstratives. Elbourne (2008) proposes a relational component for English demonstratives, which can allow for a more flexible functional relation between the index argument and the referent of the demonstrative description, thus unifying the treatment of anaphoric and relational uses.

Our studies show that the degree to which such relational extension is possible is significantly different between English and Mandarin. This may be due to English demonstratives not having a relational meaning, but as we discussed in Section 4.2, another possibility is that English demonstratives do have the relational semantics but are blocked by the definite article. Note that the idea of subsuming the anaphoric interpretation into the relational meaning has been extended to strong articles in Austro-Bavarian German in Simonenko 2014 and Simonenko 2015 and applied to bridging cases in Hanink 2017 (see also Hanink 2021). Whether this unified view of anaphoric and relational definites can be extended to all languages that have been shown to distinguish weak vs. strong definites is an open and interesting question that can be investigated in the future. Our proposal of Mandarin *na* as a relational demonstrative adds to this literature another language that has a weak and strong distinction in definiteness and allows a generally relational meaning for the strong definite.

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Appendix

Study 4 Relational Ratings Task (Mandarin): Additional plots

Fig. 16 through 20 plot the ratings in each control condition.

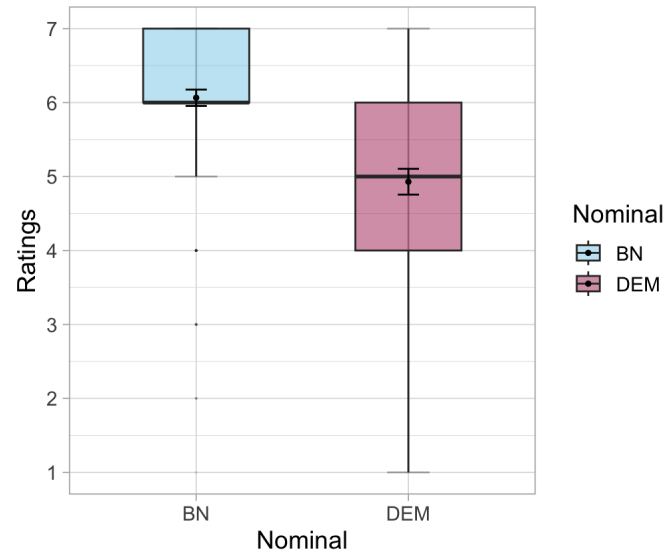


Figure 16: Ratings as function of nominal type (color/bottom) in BOTHGOOD

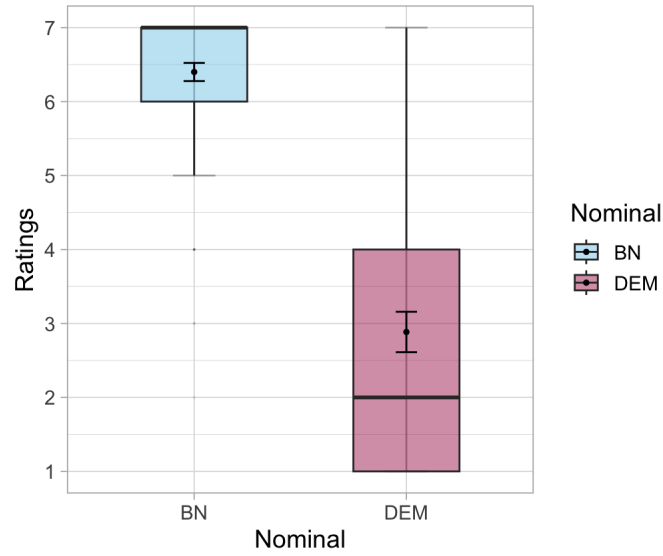


Figure 17: Ratings as function of nominal type (color/bottom) in BNGOOD_GENERIC

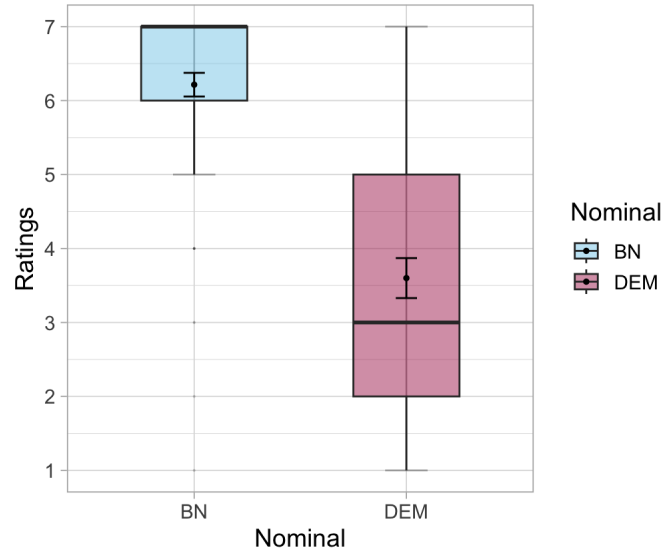


Figure 18: Ratings as function of nominal type (color/bottom) in BNGOOD_UNIQUE

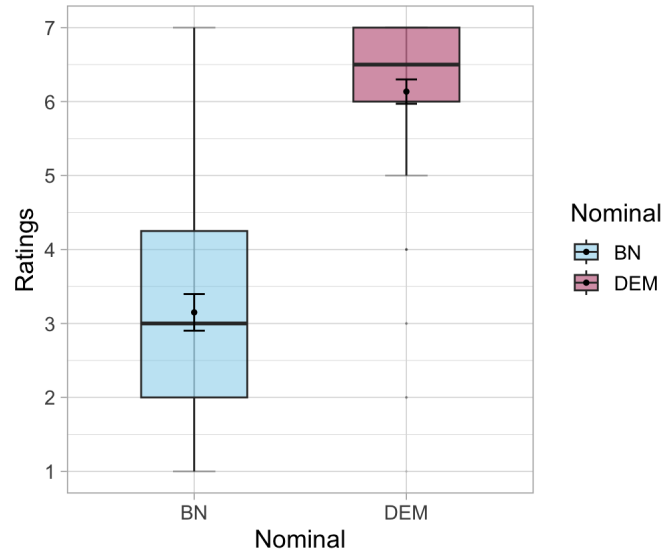


Figure 19: Ratings as function of nominal type (color/bottom) in DEMGOOD_ANTIUNIQUE

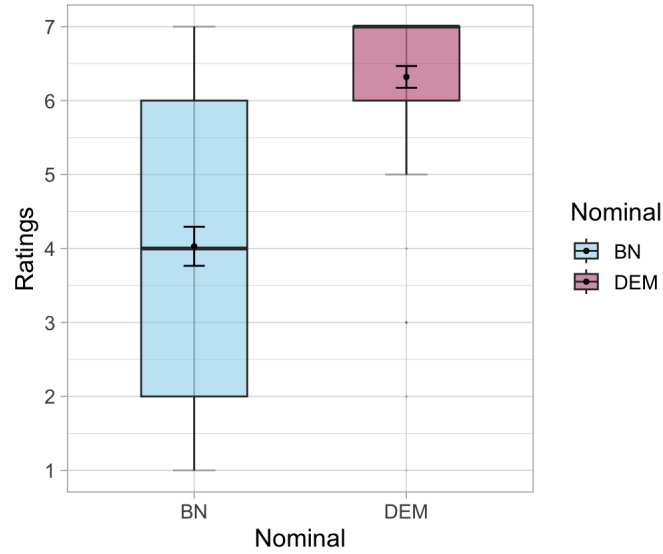


Figure 20: Ratings as function of nominal type (color/bottom) in DEMGOOD_NOUNMODIFIER